1. Write Java program to print “Hello World”

2. Write a program in java that finds second largest element in an array

3. Given three numbers, write a Java program to read three numbers from keyword and print out the largest of them.

4. Write a Java program reads a character and check if it is alphabet or not

5. Write a program that checks if the array is sorted or not

6. Write a Java program to read two integer values m and n and to decide whether m is a multiple of n.

7. Write a Java program that reads radius of circle and finds area and circumference.

8. Write a Java program that finds factorial of a positive number using recursive method

9. Write Java program to print prime numbers from 300 to 500 using method

10. Write a Java program to find the largest number among four different numbers using conditional operator

11. A non-empty array A of length n is called on array of all possibilities if it contains all numbers between 0 and A.length-1 inclusive. Write a method named is All Possibilities that accepts an integer array and returns 1 if the array is an array of all pssiblities, otherwise it returns 0.

12. Write a Java program that finds sum of two and three numbers using concept of method overloading

13. Write a Java program that finds areas of different geometric shapes using concept of method overloading.

14. Create a class Number with three int instance variable x , y and z. The class will have one constructor. The class also will contain member function getMax () that will return the largest number. Create a main method that will create an object of Number and will print the largest number.

15. Write a Java program to add two complex numbers

16. Write a Java program to add two Time(hr,min,sec) objects

17. Create a class Swapper class with two integer instance variable x and y and constructor with two parameters that initializes the two variables. Also include three member functions: A getX () that returns x, a getY () function that returns y, a void swap) method that swaps the values of x and y. Then define a main() method to create an object of Swapper class and swap the value of instance variables

18. Create a class Date with three integer instance variables named day, month, year. It has a constructor with three parameters for initializing the instance variables, and it has one-member function named daySinceJan1 (). It computes and returns the number of days since January 1 of the same year, including January 1 and the day in the Date object. For example, if day is a Date object with day = 1, month = 3 and year = 2000, then the call date.daySinceJan1 should return 61 since there are 61 days between the dates of January 1, 2000, and March 1, 2000, including January 1 and March 1. Then define main () method to handle Date class. Don’t forget leap years.

19. Create a USMoney class with two integer instance variables dollars and cents. Add a constructor with two parameters for initializing a USMoney object. The constructor should check that the cent value is between 0 and 99 and, if not, transfer some centsto the dollars variables to make it between 0 and 99. For example, if x is a USMoney object with 5 dollars and 80 cents, and if y is a USMoney object with 1 dollar and 90 cents, then x.plus (y) will return a new USMoney object with 7 dollars and 70 cents. Also, create a main () method that creates to objects of USMoney class and add them.

20. Create a Person class with private instance variables for person’s name and birth date. Add appropriate functions for these variables. Then create a subclass CollegeGraduate with private instance variables for the student’s GPA and year of graduation and appropriate functions for

these variables. Don’t forget to include appropriate constructor constructors for your classes. Then define main () method that demonstrates your classes.

21. Create a class Box with fields width, height and depth. Add methods getArea () and getVolume (). Use suitable constructors. From main () method create an object of Box class and find its area as volume.

22. Create a class Room with instance variables length and breadth. Add one function getArea () that returns the area of the room. Create a subclass MyRoom and add one instance variable height. Add one function getVolume () that returns the volume. Then define main () method that creates two MyRoom objects and find area and volumes of both rooms.

23. Create a class Box with instance variables length, breadth and height. Add one method getVolume () to compute the volume of box. Use suitable constructors. Create a subclass BoxWeight that extends Box that add one variable weight. Add one function getWeight () that displays the weight of box to this class. Add suitable constructors. Create one more subclass class Shipment that extends BoxWeight. Add one function getCost () that displays the cost of the box. Add suitable constructors. Then define main () method that creates an object of Shipment that initializes the instance variables through constructor.

24. Create an abstract class Figure with two instance variables dim1 and dim2. Add suitable constructors. Add one abstract function called getArea (). Create two subclass called Rectangle and Triangle. Add function getArea () to both of the classes that wil find the area of respective figures. Then define main () method that creates an object of each classes and find the area of triangle and rectangle.

25. Write a Java program to demonstrate divide by zero exception.

26. Write a Java program to demonstrate array index bounds exception

27. Write a Java Program to demonstrate null exception

28. Write a Java program to demonstrate custom exception

29. Write a Java program to reads contents of a file using character stream

30. Write a Java program to write some lines of text to file using character stream

31. Write a Java program that reads contents of same file using character stream

32. Write a Java program that writes line of text to file using byte stream

33. Write a Java program that reads contents of file using byte stream

34. Write a Java program to write objects to file

35. Write a Java program that creates a class called Stack and then implements push() and pop() operations

36. Create an interface Exam with methods setExam(String division, int mark) and showExam(), create a class named test that implements the interface Exam and then display the records.

37. Write a Java program to create a class Mobile (type, phone_no). Customize the exception such that if the user give phone_no having less than or greater than 10 digit, then the program has to throw an exception with message “Invalid Phone Number”

38. Create a class named Movie (id, genre). Write the object of Movie class into file named “Comedy.dat” having comedy as genre

39. Write a program to create a class student with data member roll and name. sort the 10 objects of this class on the basis of name.

40. Create a class named Book with instance variables tile and price. Add a method named setVar to pass parameters for title and price. Add another method named showVar to display values of these variables. Now in main(), declare 4 objects of book and displa the records of book that starts with “Java”.

41. Create a Shape interface having methods area() and perimeter(). Create two subclasses, Circle and Rectangle that implements the Shape interface. Create a class Sample with main

method and demonstrate the area and perimeters of both the Shape classes. You need to handle the values of length, breadth and radius in respective classes to calculate their area and perimeter.

42. Create a class Student with private member variables name and percentage. Write methods to set, display and return values of private variables in the Student class. Create 10 different objects of the student class, set the values, ad display name of Stuent who have highest average_marks in the main method of another class named StudentDemo

43. Write a program to illustrate the concept of ArrayIndexOutOfBoundsException

44. Write a Simple GUI program that displays "hello World" in a text field. The program should display if user clicks a button.

45. Write GUI program using Swing components to find sum and difference of two numbers. Use two text fields for giving input and a label for output. The program should display sum if user presses mouse and difference if user release mouse.

46. You are hired by a reputed software company which is going to design an application for "Movie Rental System". Your responsibility is to design a schema named MRS and create a table named Movie(id, Title, Genre, Language, Length). Write a program to design a GUI to take input for this table and insert the data into table after clicking OK button

47. Write an applet that reads two numbers from HTML file as parameter and displays the sum of these two numbers.

48. Write a Java program in awt to create form to enter employee information (eid, ename, salary, gender)

49. Write a Java program to demonstrate FlowLayout

50. Write a Java Program to demonstrate GridLayout

51. Write a program using swing components to add two numbers. Use text fields for inputs and output. Your program should display the result when the user presses a button.

52. Write a Java program who live in Kathmandu district, assuming that the student table has four attributes (ID, name, district and age).

53. Write a Java program to insert one record to database. Assume your own database and table

54. Write a Java Program to delete a record from database. Assume your own database and table

55. Create a servlet that displays two text boxes in web browser, reads number entered in first text box, calculates factorial and displays it in second textfield.

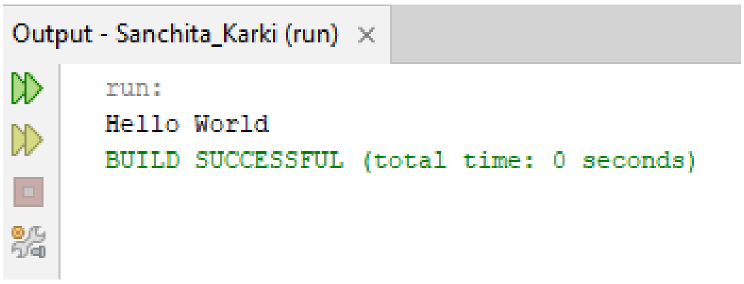
56. Write a servlet program that reads two numbers from web browser and finds sum of these two numbers.

57. Write a JSP program display text "Apache Tomcat" 10 times.

58. How exceptions can be handled in JSP scripts? Explain with suitable JSP script

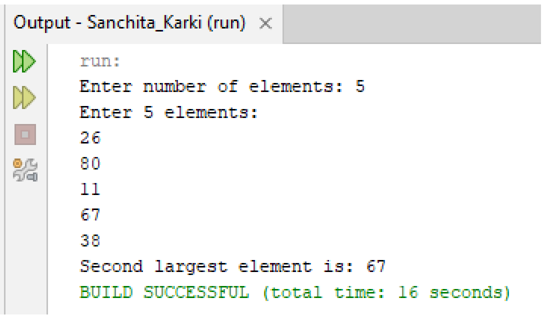
1. Write Java program to print “Hello World”

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```



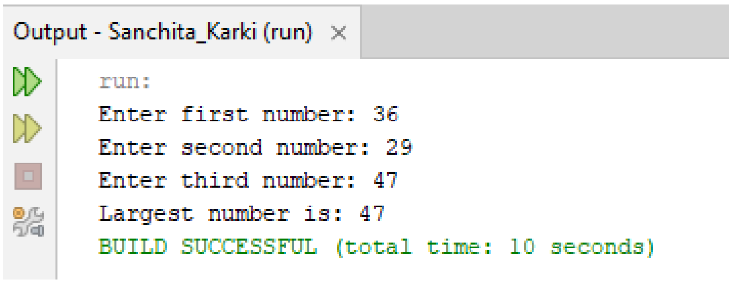
2. Write a program in java that finds second largest element in an array

```java
import java.util.Scanner;
public class SecondLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        if (n < 2) {
            System.out.println("Array must contain at least two elements.");
            return;
        }
        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;
        for (int i = 0; i < n; i++) {
            if (arr[i] > largest) {
                secondLargest = largest;
                largest = arr[i];
            } else if (arr[i] > secondLargest && arr[i] != largest) {
                secondLargest = arr[i];
            }
        }
        if (secondLargest == Integer.MIN_VALUE) {
            System.out.println("No distinct second largest element found.");
        } else {
            System.out.println("Second largest element is: " + secondLargest);
        }
        sc.close();
    }
}
```



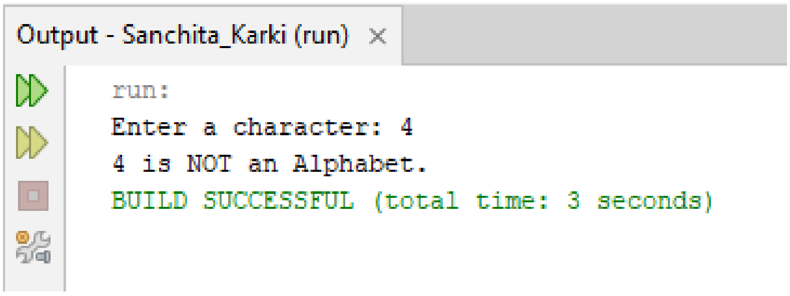
3. Given three numbers, write a Java program to read three numbers from keyword and print out the largest of them.

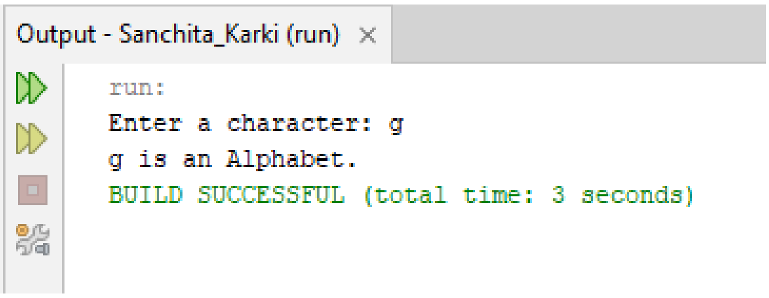
```
import java.util.Scanner;
public class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.print("Enter third number: ");
        int c = sc.nextInt();
        int largest;
        if (a >= b && a >= c) {
            largest = a;
        } else if (b >= a && b >= c) {
            largest = b;
        } else {
            largest = c;
        }
        System.out.println("Largest number is: " + largest);
        sc.close();
    }
}
```



4. Write a Java program reads a character and check if it is alphabet or not

```java
import java.util.Scanner;
public class CheckAlphabet {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
            System.out.println(ch + " is an Alphabet.");
        } else {
            System.out.println(ch + " is NOT an Alphabet.");
        }
        sc.close();
    }
}
```





5. Write a program that checks if the array is sorted or not

```java
import java.util.Scanner;
public class CheckSortedArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        boolean isSorted = true;
        for (int i = 0; i < n - 1; i++) {
            if (arr[i] > arr[i + 1]) {
                isSorted = false;
                break;
            }
        }
        if (isSorted) {
            System.out.println("Array is sorted in ascending order.");
        } else {
            System.out.println("Array is NOT sorted in ascending order.");
        }
        sc.close();
    }
}
```

Output - Sanchita_Karki (run)

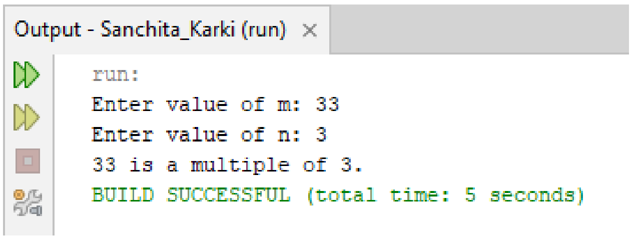
```
run:
Enter number of elements: 4
Enter 4 elements:
15
26
33
45
Array is sorted in ascending order.
BUILD SUCCESSFUL (total time: 10 seconds)
```

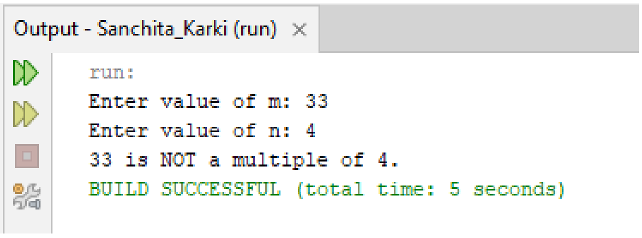
Output - Sanchita_Karki (run)

```
run:
Enter number of elements: 5
Enter 5 elements:
29
45
12
57
67
Array is NOT sorted in ascending order.
BUILD SUCCESSFUL (total time: 13 seconds)
```

6. Write a Java program to read two integer values m and n and to decide whether m is a multiple of n.

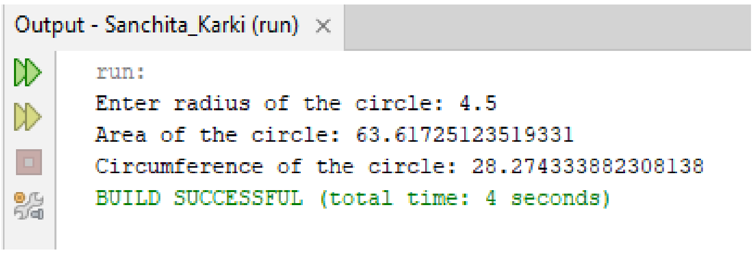
```
import java.util.Scanner;
public class MultipleCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter value of m: ");
        int m = sc.nextInt();
        System.out.print("Enter value of n: ");
        int n = sc.nextInt();
        if (n == 0) {
            System.out.println("Cannot check multiple because division by zero is not allowed.");
        } else if (m % n == 0) {
            System.out.println(m + " is a multiple of " + n + ".");
        } else {
            System.out.println(m + " is NOT a multiple of " + n + ".");
        }
        sc.close();
    }
}
```





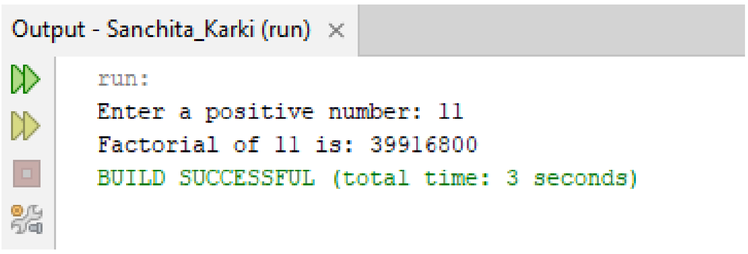
7. Write a Java program that reads radius of circle and finds area and circumference.

```
import java.util.Scanner;
public class CircleCalculation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter radius of the circle: ");
        double radius = sc.nextDouble();
        double area = Math.PI * radius * radius;
        double circumference = 2 * Math.PI * radius;
        System.out.println("Area of the circle: " + area);
        System.out.println("Circumference of the circle: " + circumference);
        sc.close();
    }
}
```



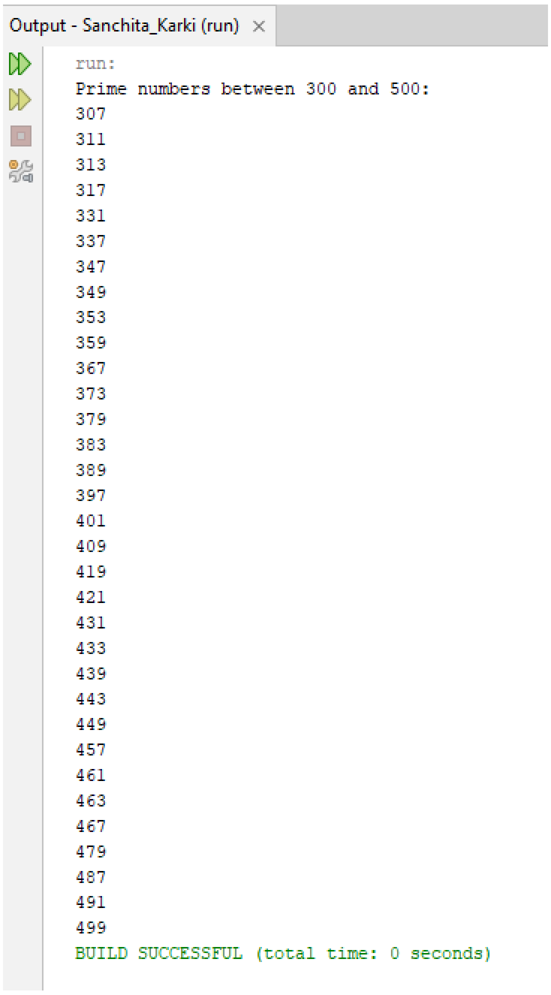
8. Write a Java program that finds factorial of a positive number using recursive method

```
import java.util.Scanner;
public class FactorialRecursive {
   static int factorial(int n) {
      if (n == 0 || n == 1) {
         return 1;
      }
      return n * factorial(n - 1);
   }
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter a positive number: ");
      int num = sc.nextInt();
      if (num < 0) {
         System.out.println("Factorial is not defined for negative numbers.");
      } else {
         System.out.println("Factorial of " + num + " is: " + factorial(num));
      }
      sc.close();
   }
}
```



9. Write Java program to print prime numbers from 300 to 500 using method

```java
public class PrimeNumbers {
    static boolean isPrime(int num) {
        if (num < 2) {
            return false;
        }
        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0) {
                return false;
            }
        }
        return true;
    }
    public static void main(String[] args) {
        System.out.println("Prime numbers between 300 and 500:");

        for (int i = 300; i <= 500; i++) {
            if (isPrime(i)) {
                System.out.println(i);
            }
        }
    }
}
```

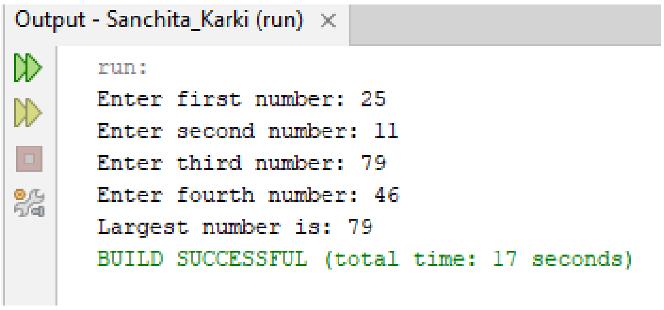


Output - Sanchita_Karki (run)

```
run:
Prime numbers between 300 and 500:
307
311
313
317
331
337
347
349
353
359
367
373
379
383
389
397
401
409
419
421
431
433
439
443
449
457
461
463
467
479
487
491
499
BUILD SUCCESSFUL (total time: 0 seconds)
```

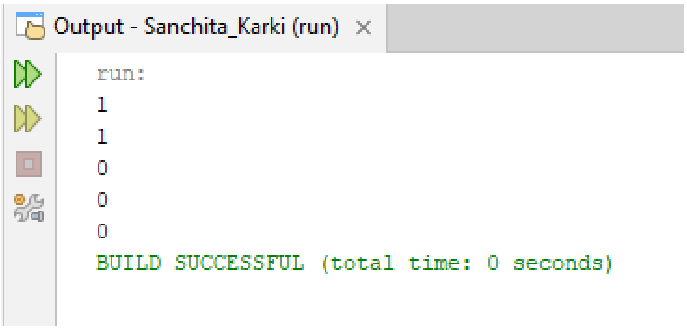
10. Write a Java program to find the largest number among four different numbers using conditional operator

```java
import java.util.Scanner;
public class LargestOfFour {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.print("Enter third number: ");
        int c = sc.nextInt();
        System.out.print("Enter fourth number: ");
        int d = sc.nextInt();
        int largest = (a > b) ? ((a > c) ? ((a > d) ? a : d) : ((c > d) ? c : d))
                    : ((b > c) ? ((b > d) ? b : d) : ((c > d) ? c : d));
        System.out.println("Largest number is: " + largest);
        sc.close();
    }
}
```



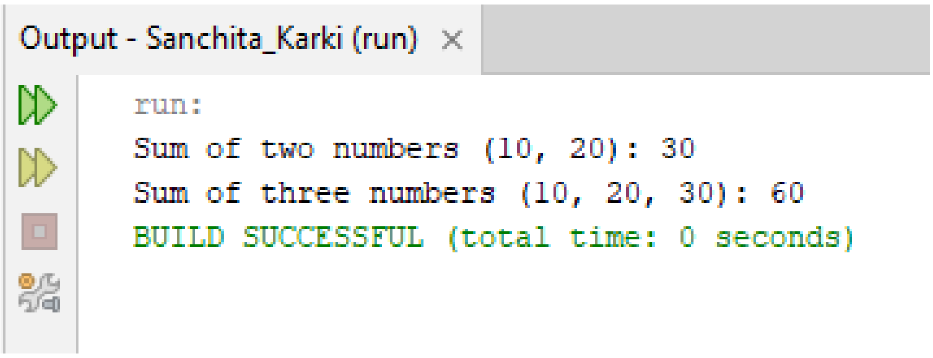
11. A non-empty array A of length n is called on array of all possibilities if it contains all numbers between 0 and A.length-1 inclusive. Write a method named is All Possibilities that accepts an integer array and returns 1 if the array is an array of all possiblities, otherwise it returns 0.

```
public class AllPossibilities {
    public static int isAllPossibilities(int[] A) {
        int n = A.length;
        if (n == 0) return 0;
        boolean[] found = new boolean[n];
        for (int i = 0; i < n; i++) {
            int val = A[i];
            if (val < 0 || val >= n) {
                return 0;
            }
            found[val] = true;
        }
        for (int i = 0; i < n; i++) {
            if (!found[i]) {
                return 0;
            }
        }
        return 1;
    }
    public static void main(String[] args) {
        int[] A1 = {1, 0, 2, 3};
        int[] A2 = {3, 1, 2, 0};
        int[] A3 = {1, 2, 3, 4};
        int[] A4 = {0, 1, 2, 2};
        int[] A5 = {};
        System.out.println(isAllPossibilities(A1));
        System.out.println(isAllPossibilities(A2));
        System.out.println(isAllPossibilities(A3));
        System.out.println(isAllPossibilities(A4));
        System.out.println(isAllPossibilities(A5));
    }
}
```



12. Write a Java program that finds sum of two and three numbers using concept of method overloading

```
public class SumOverloading {
    static int sum(int a, int b) {
        return a + b;
    }
    static int sum(int a, int b, int c) {
        return a + b + c;
    }
    public static void main(String[] args) {
        System.out.println("Sum of two numbers (10, 20): " + sum(10, 20));
        System.out.println("Sum of three numbers (10, 20, 30): " + sum(10, 20, 30));
    }
}
```



13. Write a Java program that finds areas of different geometric shapes using concept of method overloading.

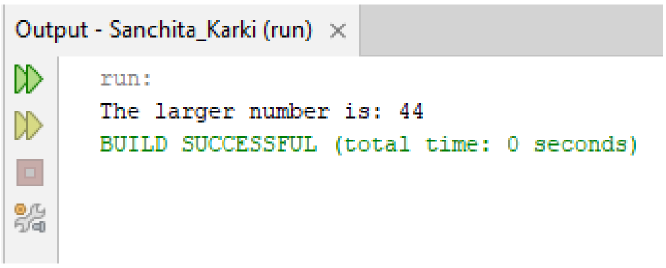
```java
import java.util.Scanner;
public class AreaOverloading {
    static double area(double side) {
        return side * side;
    }
    static double area(double length, double width) {
        return length * width;
    }
    static double area(int radius) {
        return Math.PI * radius * radius;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter side of square: ");
        double side = sc.nextDouble();
        System.out.println("Area of square: " + area(side));
        System.out.print("Enter length and width of rectangle: ");
        double length = sc.nextDouble();
        double width = sc.nextDouble();
        System.out.println("Area of rectangle: " + area(length, width));
        System.out.print("Enter radius of circle: ");
        int radius = sc.nextInt();
        System.out.println("Area of circle: " + area(radius));
        sc.close();
    }
}
```

Output - Sanchita_Karki (run)

```
run:
Enter side of square: 5
Area of square: 25.0
Enter length and width of rectangle: 12.5
6
Area of rectangle: 75.0
Enter radius of circle: 4
Area of circle: 50.26548245743669
BUILD SUCCESSFUL (total time: 20 seconds)
```

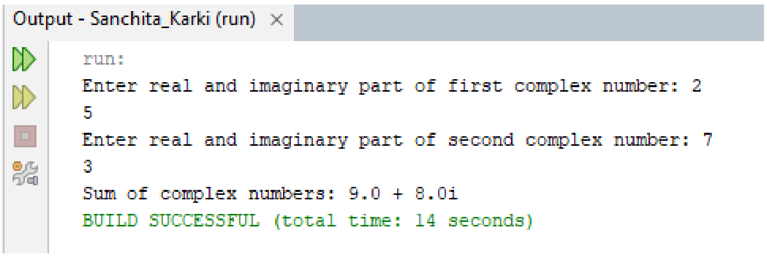
14. Create a class Number with three int instance variable x , y and z. The class will have one constructor. The class also will contain member function getMax () that will return the largest number. Create a main method that will create an object of Number and will print the largest number.

```
class Number {
    int x;
    int y;
    int z;
    Number(int x, int y, int z) {
        this.x = x;
        this.y = y;
        this.z = z;
    }
    int getMax() {
        int max = x;
        if (y > max) {
            max = y;
        }
        if (z > max) {
            max = z;
        }
        return max;
    }
}
public class ThreeNumbers {
    public static void main(String[] args) {
        Number obj = new Number(25, 44, 12);
        System.out.println("The larger number is: " + obj.getMax());
    }
}
```



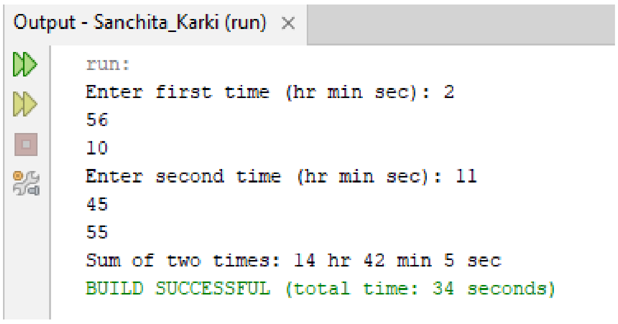
15. Write a Java program to add two complex numbers

```
import java.util.Scanner;
class Complex {
    double real;
    double imag;
    Complex(double real, double imag) {
        this.real = real;
        this.imag = imag;
    }
    Complex add(Complex c) {
        return new Complex(this.real + c.real, this.imag + c.imag);
    }
    void display() {
        if (imag >= 0)
            System.out.println(real + " + " + imag + "i");
        else
            System.out.println(real + " - " + (-imag) + "i");
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter real and imaginary part of first complex number: ");
        double r1 = sc.nextDouble();
        double i1 = sc.nextDouble();
        System.out.print("Enter real and imaginary part of second complex number: ");
        double r2 = sc.nextDouble();
        double i2 = sc.nextDouble();
        Complex c1 = new Complex(r1, i1);
        Complex c2 = new Complex(r2, i2);
        Complex result = c1.add(c2);
        System.out.print("Sum of complex numbers: ");
        result.display();
        sc.close();
    }
}
```



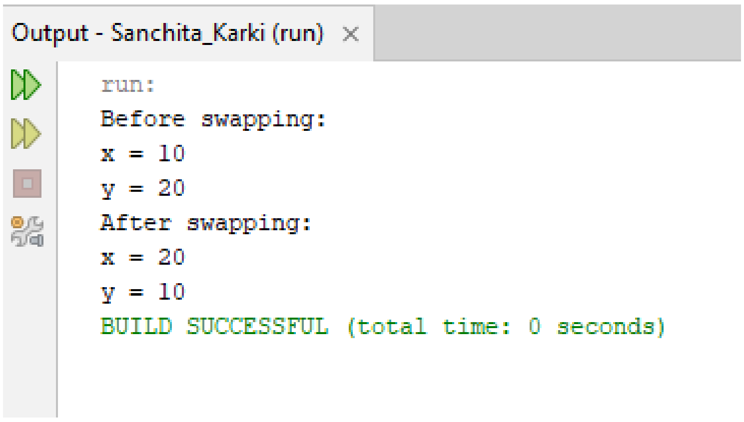
16. Write a Java program to add two Time(hr,min,sec) objects

```java
import java.util.Scanner;
class Time {
    int hr, min, sec;
    Time(int hr, int min, int sec) {
        this.hr = hr;
        this.min = min;
        this.sec = sec;
    }
    Time add(Time t) {
        int newSec = this.sec + t.sec;
        int newMin = this.min + t.min;
        int newHr = this.hr + t.hr;
        if (newSec >= 60) {
            newMin += newSec / 60;
            newSec %= 60;
        }
        if (newMin >= 60) {
            newHr += newMin / 60;
            newMin %= 60;
        }
        return new Time(newHr, newMin, newSec);
    }
    void display() {
        System.out.println(hr + " hr " + min + " min " + sec + " sec");
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first time (hr min sec): ");
        Time t1 = new Time(sc.nextInt(), sc.nextInt(), sc.nextInt());
        System.out.print("Enter second time (hr min sec): ");
        Time t2 = new Time(sc.nextInt(), sc.nextInt(), sc.nextInt());
        Time result = t1.add(t2);
        System.out.print("Sum of two times: ");
        result.display();
        sc.close();
    }
}
```



17. Create a class Swapper class with two integer instance variable x and y and constructor with two parameters that initializes the two variables. Also include three member functions: A getX () that returns x, a getY () function that returns y, a void swap () method that swaps the values of x and y. Then define a main() method to create an object of Swapper class and swap the value of instance variables

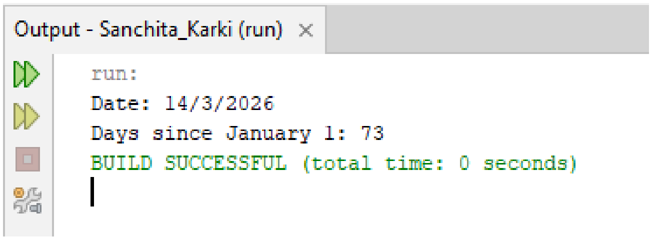
```
class Swapper {
    int x;
    int y;
    Swapper(int x, int y) {
        this.x = x;
        this.y = y;
    }
    int getX() {
        return x;
    }
    int getY() {
        return y;
    }
    void swap() {
        int temp = x;
        x = y;
        y = temp;
    }
    public static void main(String[] args) {
        Swapper obj = new Swapper(10, 20);
        System.out.println("Before swapping:");
        System.out.println("x = " + obj.getX());
        System.out.println("y = " + obj.getY());
        obj.swap();
        System.out.println("After swapping:");
        System.out.println("x = " + obj.getX());
        System.out.println("y = " + obj.getY());
    }
}
```


Output - Sanchita_Karki (run)

```
run:
Before swapping:
x = 10
y = 20
After swapping:
x = 20
y = 10
BUILD SUCCESSFUL (total time: 0 seconds)
```

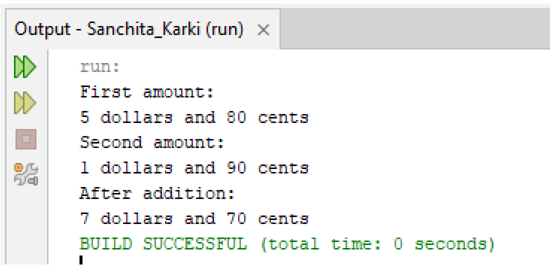
18. Create a class Date with three integer instance variables named day, month, year. It has a constructor with three parameters for initializing the instance variables, and it has one-member function named daySinceJan1 (). It computes and returns the number of days since January 1 of the same year, including January 1 and the day in the Date object. For example, if day is a Date object with day = 1, month = 3 and year = 2000, then the call date.daySinceJan1 should return 61 since there are 61 days between the dates of January 1, 2000, and March 1, 2000, including January 1 and March 1. Then define main () method to handle Date class. Don't forget leap years.

```
class Date {
    int day;
    int month;
    int year;
    Date(int day, int month, int year) {
        this.day = day;
        this.month = month;
        this.year = year;
    }
    boolean isLeapYear() {
        if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
            return true;
        }
        return false;
    }
    int daySinceJan1() {
        int[] daysInMonth = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
        if (isLeapYear()) {
            daysInMonth[1] = 29;
        }
        int totalDays = 0;
        for (int i = 0; i < month - 1; i++) {
            totalDays += daysInMonth[i];
        }
        totalDays += day;
        return totalDays;
    }
    public static void main(String[] args) {
        Date date = new Date(1, 3, 2000);
        System.out.println("Date: " + date.day + "/" + date.month + "/" + date.year);
        System.out.println("Days since January 1: " + date.daySinceJan1());
    }
}
```



19. Create a USMoney class with two integer instance variables dollars and cents. Add a constructor with two parameters for initializing a USMoney object. The constructor should check that the cent value is between 0 and 99 and, if not, transfer some cents to the dollars variables to make it between 0 and 99. For example, if x is a USMoney object with 5 dollars and 80 cents, and if y is a USMoney object with 1 dollar and 90 cents, then x.plus (y) will return a new USMoney object with 7 dollars and 70 cents. Also, create a main () method that creates to objects of USMoney class and add them.

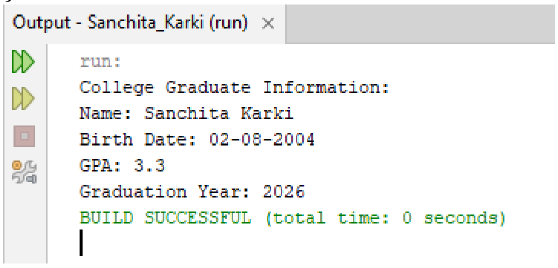
```
class USMoney {
   int dollars;
   int cents;
   USMoney(int dollars, int cents) {
      this.dollars = dollars;
      this.cents = cents;
      if (this.cents >= 100 || this.cents < 0) {
         this.dollars += this.cents / 100;
         this.cents = this.cents % 100;
         if (this.cents < 0) {
            this.cents += 100;
            this.dollars -= 1;
         }
      }
   }
   USMoney plus(USMoney other) {
      int newDollars = this.dollars + other.dollars;
      int newCents = this.cents + other.cents;

      return new USMoney(newDollars, newCents);
   }
   void display() {
      System.out.println(dollars + " dollars and " + cents + " cents");
   }
   public static void main(String[] args) {
      USMoney x = new USMoney(5, 80);
      USMoney y = new USMoney(1, 90);

      USMoney result = x.plus(y);
      System.out.println("First amount:");
      x.display();
      System.out.println("Second amount:");
      y.display();
      System.out.println("After addition:");
      result.display();
   }
}
```



20. Create a Person class with private instance variables for person's name and birth date. Add appropriate functions for these variables. Then create a subclass CollegeGraduate with private instance variables for the student's GPA and year of graduation and appropriate functions for these variables. Don't forget to include appropriate constructor constructors for your classes. Then define main () method that demonstrates your classes.

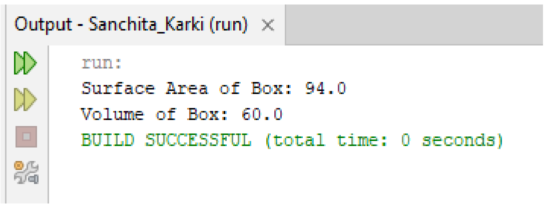
```
class Person {
   private String name;
   private String birthDate;
   Person(String name, String birthDate) {
      this.name = name;
      this.birthDate = birthDate;
   }
   public String getName() {
      return name;
   }
   public String getBirthDate() {
      return birthDate;
   }
   public void setName(String name) {
      this.name = name;
   }
   public void setBirthDate(String birthDate) {
      this.birthDate = birthDate;
   }
   public void displayPerson() {
      System.out.println("Name: " + name);
      System.out.println("Birth Date: " + birthDate);
   }
}
class CollegeGraduate extends Person {
   private double gpa;
   private int graduationYear;
   CollegeGraduate(String name, String birthDate, double gpa, int graduationYear) {
      super(name, birthDate);
      this.gpa = gpa;
      this.graduationYear = graduationYear;
   }
   public double getGpa() {
      return gpa;
   }
   public int getGraduationYear() {
      return graduationYear;
   }
   public void setGpa(double gpa) {
      this.gpa = gpa;
   }
   public void setGraduationYear(int graduationYear) {
      this.graduationYear = graduationYear;
   }
   public void displayGraduate() {
      displayPerson();
      System.out.println("GPA: " + gpa);
      System.out.println("Graduation Year: " + graduationYear);
```

```
    }
}
public class College {
    public static void main(String[] args) {
        CollegeGraduate student = new CollegeGraduate(
                "Sanchita Karki",
                "02-08-2004",
                3.3,
                2026
        );
        System.out.println("College Graduate Information:");
        student.displayGraduate();
    }
}
```



21. Create a class Box with fields width, height and depth. Add methods getArea () and getVolume (). Use suitable constructors. From main () method create an object of Box class and find its area as volume.

```
class Box {
    double width;
    double height;
    double depth;
    Box(double width, double height, double depth) {
        this.width = width;
        this.height = height;
        this.depth = depth;
    }
    double getArea() {
        return 2 * (width * height + width * depth + height * depth);
    }
    double getVolume() {
        return width * height * depth;
    }
}
public class BoxExample {
  public static void main(String[] args) {
        Box box = new Box(5, 4, 3);
        System.out.println("Surface Area of Box: " + box.getArea());
        System.out.println("Volume of Box: " + box.getVolume());
    }
}
```



22. Create a class Room with instance variables length and breadth. Add one function getArea () that returns the area of the room. Create a subclass MyRoom and add one instance variable height. Add one function getVolume () that returns the volume. Then define main () method that creates two MyRoom objects and find area and volumes of both rooms.

```
class Room {
   double length;
   double breadth;
   Room(double length, double breadth) {
      this.length = length;
      this.breadth = breadth;
   }
   double getArea() {
      return length * breadth;
   }
}
class MyRoom extends Room {
   double height;
   MyRoom(double length, double breadth, double height) {
      super(length, breadth);
      this.height = height;
   }
   double getVolume() {
      return getArea() * height;
   }
}
public class Main {
   public static void main(String[] args) {
      MyRoom room1 = new MyRoom(5, 4, 3);
      System.out.println("Room 1 Area: " + room1.getArea());
      System.out.println("Room 1 Volume: " + room1.getVolume());
      MyRoom room2 = new MyRoom(6, 5, 4);
      System.out.println("Room 2 Area: " + room2.getArea());
      System.out.println("Room 2 Volume: " + room2.getVolume());
   }
}
```

Output - Sanchita_Karki (run)

```
run:
Room 1 Area: 20.0
Room 1 Volume: 60.0
Room 2 Area: 30.0
Room 2 Volume: 120.0
BUILD SUCCESSFUL (total time: 0 seconds)
```

23. Create a class Box with instance variables length, breadth and height. Add one method getVolume () to compute the volume of box. Use suitable constructors. Create a subclass BoxWeight that extends Box that add one variable weight. Add one function getWeight () that displays the weight of box to this class. Add suitable constructors. Create one more subclass class Shipment that extends BoxWeight. Add one function getCost () that displays the cost of the box. Add suitable constructors. Then define main () method that creates an object of Shipment that initializes the instance variables through constructor.

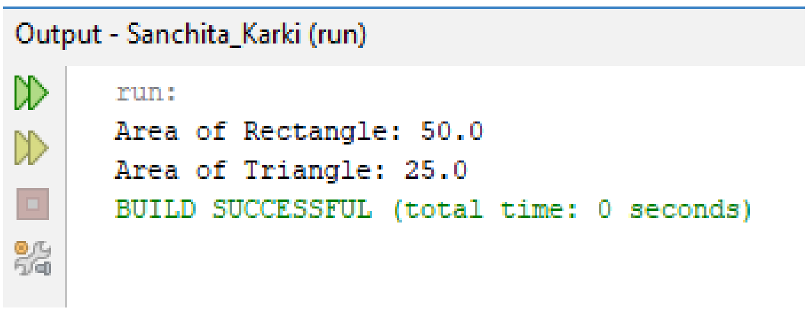
```
class Box {
   double length;
   double breadth;
   double height;
   Box(double length, double breadth, double height) {
      this.length = length;
      this.breadth = breadth;
      this.height = height;
   }
   double getVolume() {
      return length * breadth * height;
   }
}
class BoxWeight extends Box {
   double weight;
   BoxWeight(double length, double breadth, double height, double weight) {
      super(length, breadth, height);
      this.weight = weight;
   }
   void getWeight() {
      System.out.println("Weight of Box: " + weight + " kg");
   }
}
class Shipment extends BoxWeight {
   double cost;
   Shipment(double length, double breadth, double height, double weight, double cost) {
      super(length, breadth, height, weight); // Call parent constructor
      this.cost = cost;
   }
   void getCost() {
      System.out.println("Cost of Shipment: $" + cost);
   }
}
public class Main {
   public static void main(String[] args) {
      Shipment shipment = new Shipment(5, 4, 3, 12.5, 50);
      System.out.println("Volume of Box: " + shipment.getVolume());
      shipment.getWeight();
      shipment.getCost();
   }
}
```

Output - Sanchita_Karki (run)

```
run:
Volume of Box: 60.0
Weight of Box: 12.5 kg
Cost of Shipment: $50.0
BUILD SUCCESSFUL (total time: 0 seconds)
```

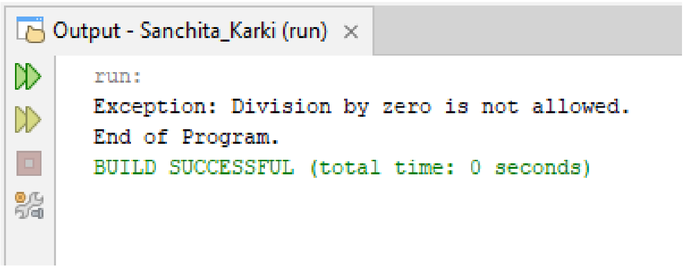
24. Create an abstract class Figure with two instance variables dim1 and dim2. Add suitable constructors. Add one abstract function called getArea (). Create two subclass called Rectangle and Triangle. Add function getArea () to both of the classes that will find the area of respective figures. Then define main () method that creates an object of each classes and find the area of triangle and rectangle.

```
abstract class Figure {
   double dim1;
   double dim2;
   Figure(double dim1, double dim2) {
      this.dim1 = dim1;
      this.dim2 = dim2;
   }
   abstract double getArea();
}
class Rectangle extends Figure {
   Rectangle(double length, double breadth) {
      super(length, breadth);
   }
   double getArea() {
      return dim1 * dim2;
   }
}
class Triangle extends Figure {
   Triangle(double base, double height) {
      super(base, height);
   }
   double getArea() {
      return 0.5 * dim1 * dim2;
   }
}
public class Main {
   public static void main(String[] args) {
      Rectangle rect = new Rectangle(10, 5);
      Triangle tri = new Triangle(10, 5);
      System.out.println("Area of Rectangle: " + rect.getArea());
      System.out.println("Area of Triangle: " + tri.getArea());
   }
}
```



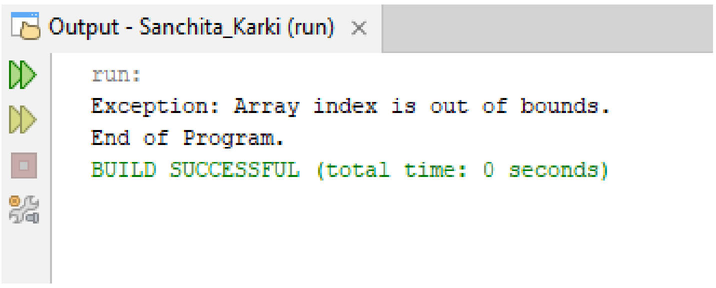
25. Write a Java program to demonstrate divide by zero exception.

```
public class DivideByZeroExceptionExample {
    public static void main(String[] args) {
        try {
            int a = 10;
            int b = 0;
            int result = a / b;
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            System.out.println("Exception: Division by zero is not allowed.");
        }
        System.out.println("End of Program.");
    }
}
```



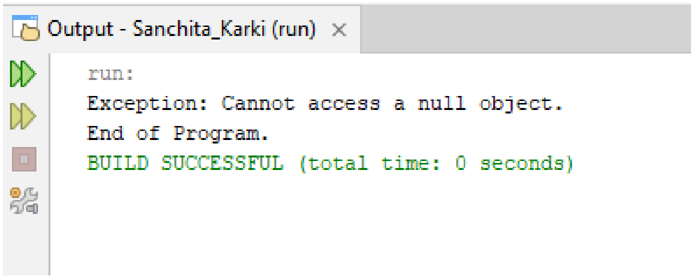
26. Write a Java program to demonstrate array index bounds exception

```java
public class ArrayIndexExceptionExample {
    public static void main(String[] args) {
        try {
            int[] arr = {10, 20, 30, 40, 50};
            System.out.println("Element: " + arr[10]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Exception: Array index is out of bounds.");
        }
        System.out.println("End of Program.");
    }
}
```



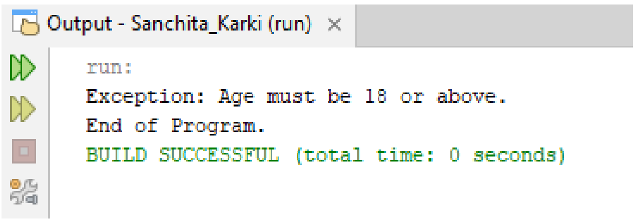
27. Write a Java Program to demonstrate null exception

```java
public class NullExceptionExample {
    public static void main(String[] args) {
        try {
            String str = null;
            System.out.println("Length: " + str.length());
        } catch (NullPointerException e) {
            System.out.println("Exception: Cannot access a null object.");
        }
        System.out.println("End of Program.");
    }
}
```



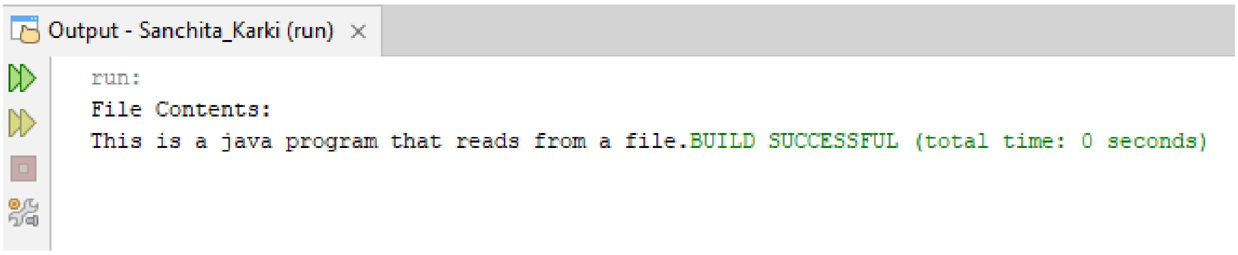
28. Write a Java program to demonstrate custom exception

```java
class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}
public class CustomExceptionExample {
    static void checkAge(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("Age must be 18 or above.");
        } else {
            System.out.println("Valid age.");
        }
    }
    public static void main(String[] args) {
        try {
            checkAge(15);
        } catch (InvalidAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }
        System.out.println("End of Program.");
    }
}
```



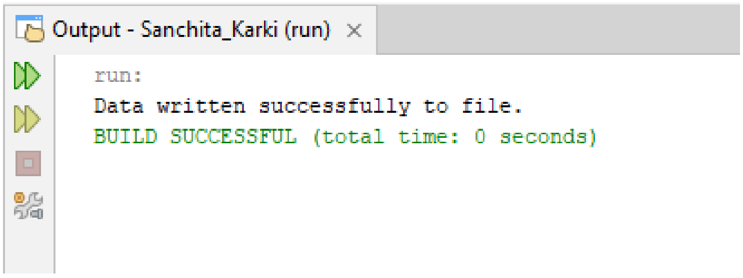
29. Write a Java program to reads contents of a file using character stream

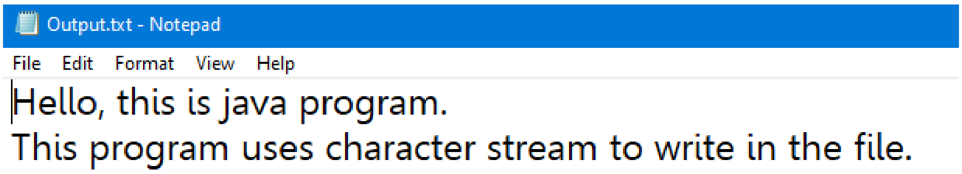
```
import java.io.FileReader;
import java.io.IOException;
public class FileReadCharacterStream {
    public static void main(String[] args) {
        try {
            FileReader fr = new FileReader("D:\\Sanchita Karki\\Input.txt");
            int ch;
            System.out.println("File Contents:");
            while ((ch = fr.read()) != -1) {
                System.out.print((char) ch);
            }
            fr.close();
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```



30. Write a Java program to write some lines of text to file using character stream

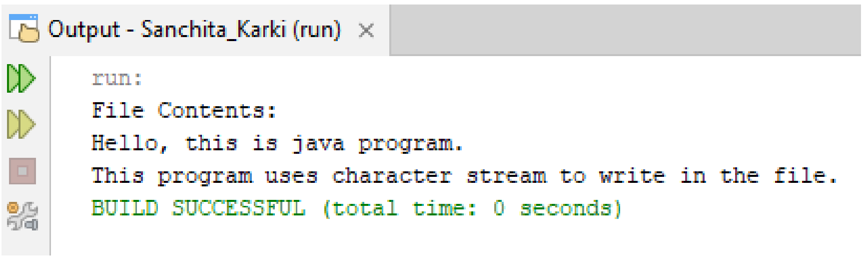
```
import java.io.FileWriter;
import java.io.IOException;
public class FileWriteCharacterStream {
    public static void main(String[] args) {
        try {
            FileWriter fw = new FileWriter("D:\\Sanchita Karki\\Output.txt");
            fw.write("Hello, this is java program.\n");
            fw.write("This program uses character stream to write in the file.\n");
            fw.close();
            System.out.println("Data written successfully to file.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```





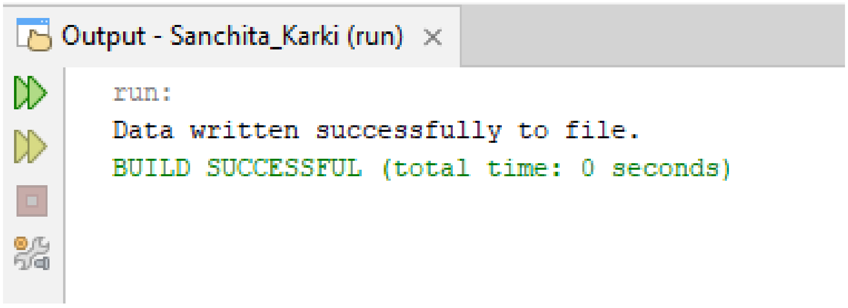
31. Write a Java program that reads contents of same file using character stream

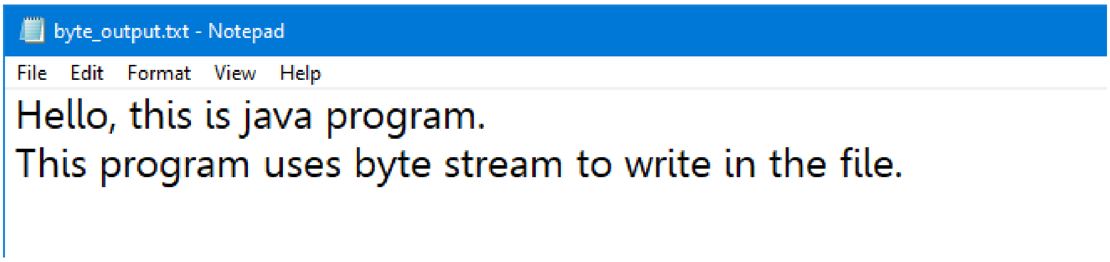
```
import java.io.FileReader;
import java.io.IOException;
public class FileReadCharacterStream {
    public static void main(String[] args) {
        try {
            FileReader fr = new FileReader("D:\\Sanchita Karki\\Output.txt");
            int ch;
            System.out.println("File Contents:");
            while ((ch = fr.read()) != -1) {
                System.out.print((char) ch);
            }
            fr.close();
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```



32. Write a Java program that writes line of text to file using byte stream

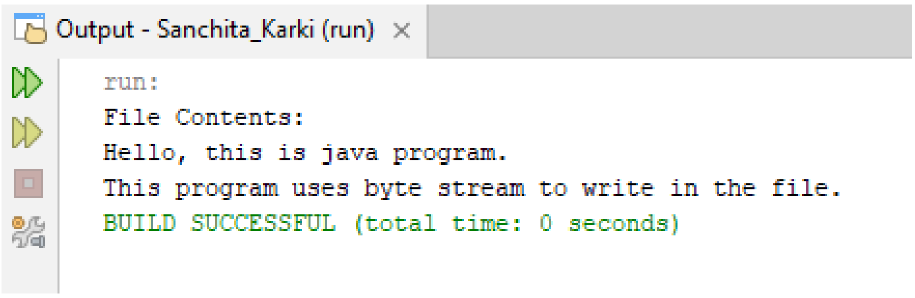
```
import java.io.FileOutputStream;
import java.io.IOException;
public class FileWriteByteStream {
    public static void main(String[] args) {
        String data = "Hello, this is java program.\n"
                  + "This program uses byte stream to write in the file.\n";
        try {
            FileOutputStream fos = new FileOutputStream("D:\\Sanchita Karki\\byte_output.txt");
            fos.write(data.getBytes());
            fos.close();
            System.out.println("Data written successfully to file.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```





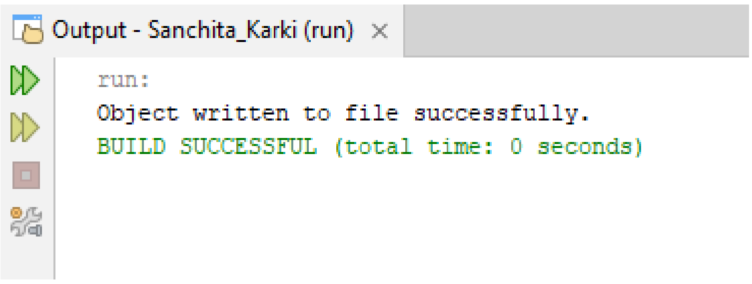
33. Write a Java program that reads contents of file using byte stream

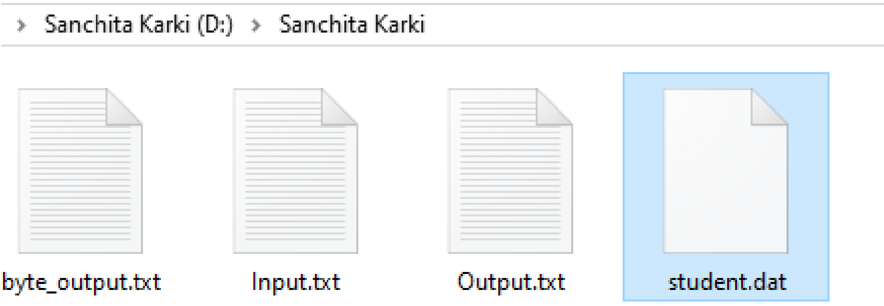
```
import java.io.FileInputStream;
import java.io.IOException;
public class FileReadByteStream {
    public static void main(String[] args) {
        try {
            FileInputStream fis = new FileInputStream("D:\\Sanchita Karki\\byte_output.txt");
            int ch;
            System.out.println("File Contents:");
            while ((ch = fis.read()) != -1) {
                System.out.print((char) ch);
            }
            fis.close(); // Close the file
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```



34. Write a Java program to write objects to file

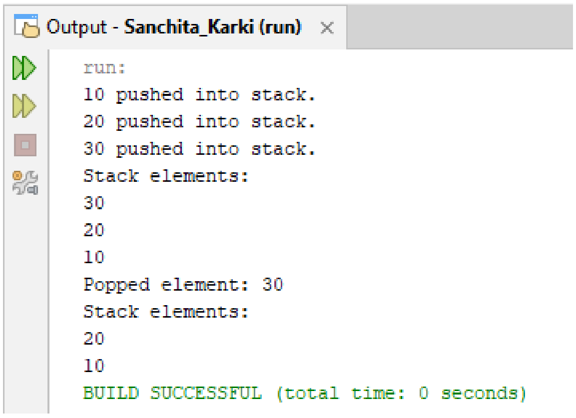
```
import java.io.FileOutputStream;
import java.io.ObjectOutputStream;
import java.io.Serializable;
import java.io.IOException;
class Student implements Serializable {
    private String name;
    private int age;
    Student(String name, int age) {
        this.name = name;
        this.age = age;
    }
    public void display() {
        System.out.println("Name: " + name + ", Age: " + age);
    }
}
public class WriteObjectToFile {
    public static void main(String[] args) {
        Student s1 = new Student("John Doe", 20);
        try {
            FileOutputStream fos = new FileOutputStream("D:\\Sanchita Karki\\student.dat");
            ObjectOutputStream oos = new ObjectOutputStream(fos);
            oos.writeObject(s1); // Writing object to file
            oos.close();
            fos.close();
            System.out.println("Object written to file successfully.");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```





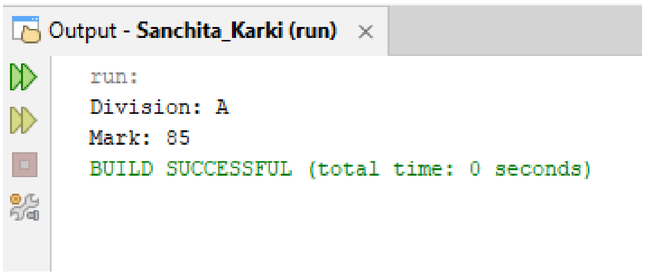
35. Write a Java program that creates a class called Stack and then implements push() and pop() operations

```
class Stack {
    private int[] stack;
    private int top;
    private int size;
    Stack(int size) {
        this.size = size;
        stack = new int[size];
        top = -1;
    }
    void push(int value) {
        if (top == size - 1) {
            System.out.println("Stack Overflow! Cannot push " + value);
        } else {
            stack[++top] = value;
            System.out.println(value + " pushed into stack.");
        }
    }
    int pop() {
        if (top == -1) {
            System.out.println("Stack Underflow! Stack is empty.");
            return -1;
        } else {
            return stack[top--];
        }
    }
    void display() {
        if (top == -1) {
            System.out.println("Stack is empty.");
        } else {
            System.out.println("Stack elements:");
            for (int i = top; i >= 0; i--) {
                System.out.println(stack[i]);
            }
        }
    }
}
public class StackExample {
    public static void main(String[] args) {
        Stack stack = new Stack(5);
        stack.push(10);
        stack.push(20);
        stack.push(30);
        stack.display();
        System.out.println("Popped element: " + stack.pop());
        stack.display();
    }
}
```


Output - Sanchita_Karki (run)
run:
10 pushed into stack.
20 pushed into stack.
30 pushed into stack.
Stack elements:
30
20
10
Popped element: 30
Stack elements:
20
10
BUILD SUCCESSFUL (total time: 0 seconds)

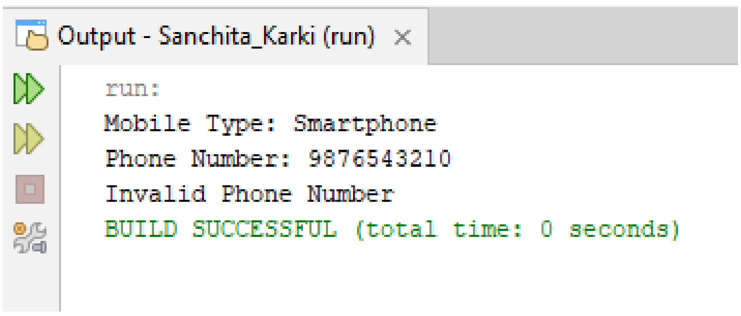
36. Create an interface Exam with methods setExam(String division, int mark) and showExam(), create a class named test that implements the interface Exam and then display the records.

```
interface Exam {
    void setExam(String division, int mark);
    void showExam();
}
class Test implements Exam {
    private String division;
    private int mark;
    public void setExam(String division, int mark) {
        this.division = division;
        this.mark = mark;
    }
    public void showExam() {
        System.out.println("Division: " + division);
        System.out.println("Mark: " + mark);
    }
}
public class InterfaceExample {
    public static void main(String[] args) {
        Test student = new Test();
        student.setExam("A", 85);
        student.showExam();
    }
}
```



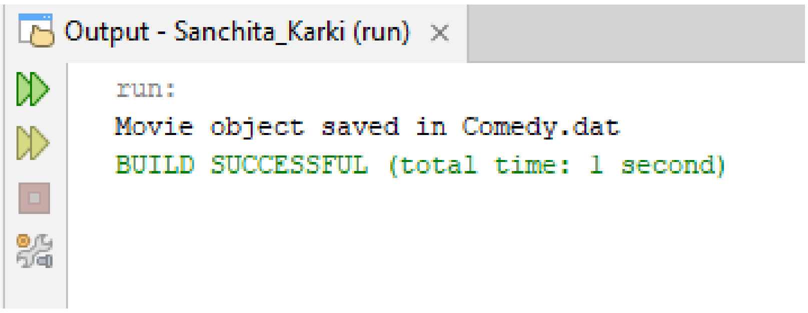
37. Write a Java program to create a class Mobile (type, phone_no). Customize the exception such that if the user give phone_no having less than or greater than 10 digit, then the program has to throw an exception with message “Invalid Phone Number”

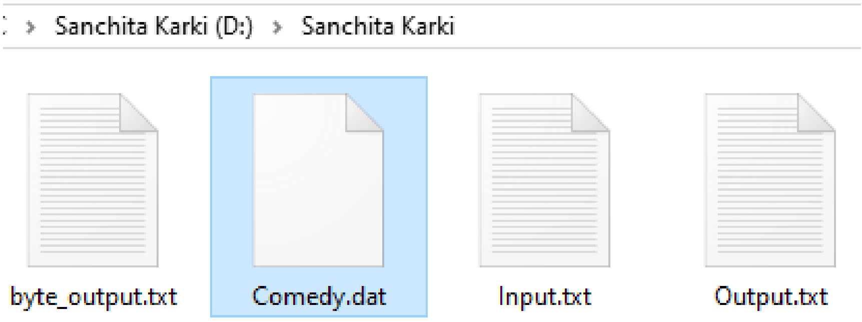
```
class InvalidPhoneException extends Exception {
    public InvalidPhoneException(String message) {
        super(message);
    }
}
class Mobile {
    private String type;
    private String phone_no;
    Mobile(String type, String phone_no) throws InvalidPhoneException {
        this.type = type;
        setPhoneNumber(phone_no);
    }
    void setPhoneNumber(String phone_no) throws InvalidPhoneException {
        if (phone_no.length() != 10 || !phone_no.matches("\\d+")) {
            throw new InvalidPhoneException("Invalid Phone Number");
        }
        this.phone_no = phone_no;
    }
    void display() {
        System.out.println("Mobile Type: " + type);
        System.out.println("Phone Number: " + phone_no);
    }
}
public class CustomizeException {
    public static void main(String[] args) {
        try {
            Mobile mobile = new Mobile("Smartphone", "9876543210");
            mobile.display();
        } catch (InvalidPhoneException e) {
            System.out.println(e.getMessage());
        }
        try {
            Mobile mobile2 = new Mobile("Smartphone", "12345"); // Invalid
            mobile2.display();
        } catch (InvalidPhoneException e) {
            System.out.println(e.getMessage());
        }
    }
}
```



38. Create a class named Movie (id, genre). Write the object of Movie class into file named “Comedy.dat” having comedy as genre

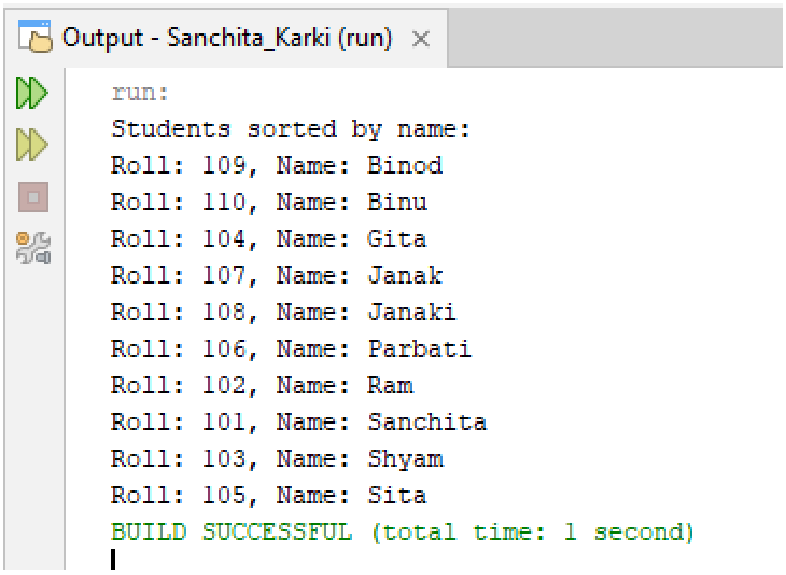
```
import java.io.FileOutputStream;
import java.io.ObjectOutputStream;
import java.io.Serializable;
import java.io.IOException;
class Movie implements Serializable {
    private int id;
    private String genre;
    Movie(int id, String genre) {
        this.id = id;
        this.genre = genre;
    }
    void saveIfComedy() {
        if (genre.equalsIgnoreCase("comedy")) {
            try {
                FileOutputStream fos = new FileOutputStream("D:\\Sanchita Karki\\Comedy.dat");
                ObjectOutputStream oos = new ObjectOutputStream(fos);
                oos.writeObject(this);
                oos.close();
                fos.close();
                System.out.println("Movie object saved in Comedy.dat");
            } catch (IOException e) {
                System.out.println("Error: " + e.getMessage());
            }
        } else {
            System.out.println("Movie genre is not comedy. Object not saved.");
        }
    }
}
public class MoveObjectToFile {
    public static void main(String[] args) {
        Movie movie = new Movie(101, "comedy");
        movie.saveIfComedy();
    }
}
```




Sanchita Karki (D:) › Sanchita Karki
byte_output.txt
Comedy.dat
Input.txt
Output.txt

39. Write a program to create a class student with data member roll and name. sort the 10 objects of this class on the basis of name.

```java
import java.util.Arrays;
import java.util.Comparator;
class Students {
    int roll;
    String name;
    Students(int roll, String name) {
        this.roll = roll;
        this.name = name;
    }
    void display() {
        System.out.println("Roll: " + roll + ", Name: " + name);
    }
}
public class SortObject {
    public static void main(String[] args) {
        Students[] students = new Students[10];
        students[0] = new Students(101, "Sanchita");
        students[1] = new Students(102, "Ram");
        students[2] = new Students(103, "Shyam");
        students[3] = new Students(104, "Gita");
        students[4] = new Students(105, "Sita");
        students[5] = new Students(106, "Parbati");
        students[6] = new Students(107, "Janak");
        students[7] = new Students(108, "Janaki");
        students[8] = new Students(109, "Binod");
        students[9] = new Students(110, "Binu");
        Arrays.sort(students, Comparator.comparing(s -> s.name));
        System.out.println("Students sorted by name:");
        for (Students s : students) {
            s.display();
        }
    }
}
```

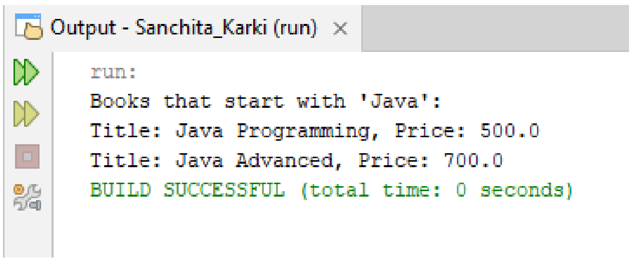


Output - Sanchita_Karki (run)

```
run:
Students sorted by name:
Roll: 109, Name: Binod
Roll: 110, Name: Binu
Roll: 104, Name: Gita
Roll: 107, Name: Janak
Roll: 108, Name: Janaki
Roll: 106, Name: Parbati
Roll: 102, Name: Ram
Roll: 101, Name: Sanchita
Roll: 103, Name: Shyam
Roll: 105, Name: Sita
BUILD SUCCESSFUL (total time: 1 second)
```

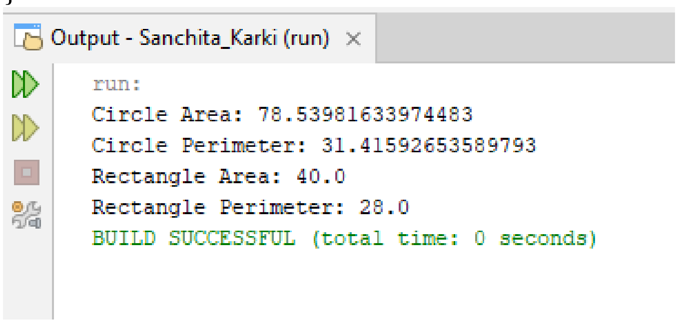
40. Create a class named Book with instance variables tile and price. Add a method named setVar to pass parameters for title and price. Add another method named showVar to display values of these variables. Now in main(), declare 4 objects of book and display the records of book that starts with “Java”.

```
class Book {
  private String title;
  private double price;
  void setVar(String title, double price) {
    this.title = title;
    this.price = price;
  }
  void showVar() {
    System.out.println("Title: " + title + ", Price: " + price);
  }
  String getTitle() {
    return title;
  }
}
public class Main {
  public static void main(String[] args) {
    Book b1 = new Book();
    Book b2 = new Book();
    Book b3 = new Book();
    Book b4 = new Book();
    b1.setVar("Java Programming", 500);
    b2.setVar("Python Basics", 400);
    b3.setVar("Java Advanced", 700);
    b4.setVar("C++ Fundamentals", 350);
    Book[] books = {b1, b2, b3, b4};
    System.out.println("Books that start with 'Java':");
    for (Book b : books) {
      if (b.getTitle().startsWith("Java")) {
        b.showVar();
      }
    }
  }
}
```



41. Create a Shape interface having methods area() and perimeter(). Create two subclasses, Circle and Rectangle that implements the Shape interface. Create a class Sample with main method and demonstrate the area and perimeters of both the Shape classes. You need to handle the values of length, breadth and radius in respective classes to calculate their area and perimeter.

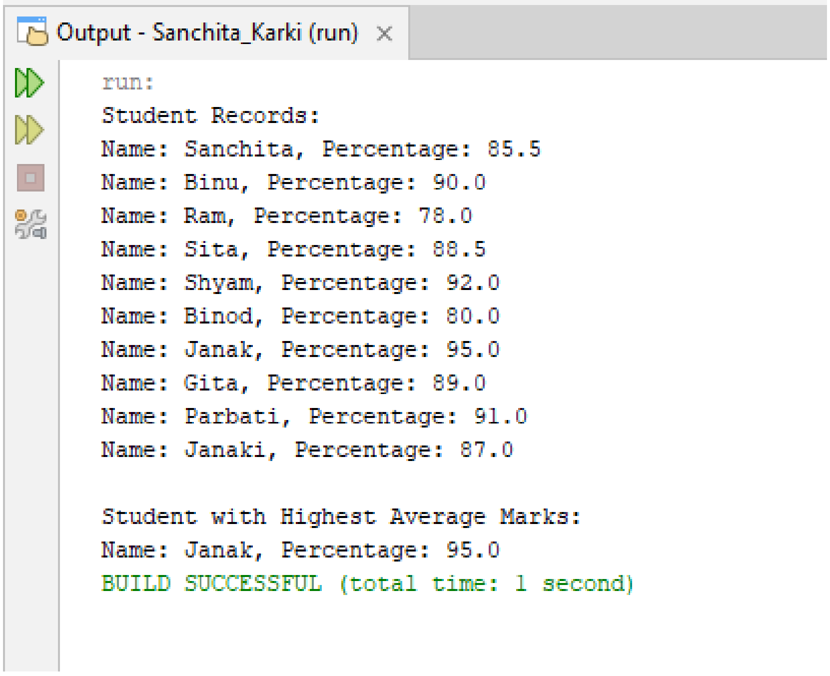
```
interface Shape {
   double area();
   double perimeter();
}
class Circle implements Shape {
   private double radius;
   Circle(double radius) {
      this.radius = radius;
   }
   public double area() {
      return Math.PI * radius * radius;
   }
   public double perimeter() {
      return 2 * Math.PI * radius;
   }
}
class Rectangle implements Shape {
   private double length;
   private double breadth;
   Rectangle(double length, double breadth) {
      this.length = length;
      this.breadth = breadth;
   }
   public double area() {
      return length * breadth;
   }
   public double perimeter() {
      return 2 * (length + breadth);
   }
}
public class Sample {
   public static void main(String[] args) {
      Shape circle = new Circle(5);
      Shape rectangle = new Rectangle(10, 4);
      System.out.println("Circle Area: " + circle.area());
      System.out.println("Circle Perimeter: " + circle.perimeter());
      System.out.println("Rectangle Area: " + rectangle.area());
      System.out.println("Rectangle Perimeter: " + rectangle.perimeter());
   }
}
```



42. Create a class Student with private member variables name and percentage. Write methods to set, display and return values of private variables in the Student class. Create 10 different objects of the student class, set the values, ad display name of Student who have highest average_marks in the main method of another class named StudentDemo

```
class Student {
    private String name;
    private double percentage;
    public void setData(String name, double percentage) {
        this.name = name;
        this.percentage = percentage;
    }
    public void display() {
        System.out.println("Name: " + name + ", Percentage: " + percentage);
    }
    public double getPercentage() {
        return percentage;
    }
    public String getName() {
        return name;
    }
}
public class StudentDemo {
    public static void main(String[] args) {
        Student[] students = new Student[10];
        students[0] = new Student();
        students[0].setData("Sanchita", 85.5);
        students[1] = new Student();
        students[1].setData("Binu", 90.0);
        students[2] = new Student();
        students[2].setData("Ram", 78.0);
        students[3] = new Student();
        students[3].setData("Sita", 88.5);
        students[4] = new Student();
        students[4].setData("Shyam", 92.0);
        students[5] = new Student();
        students[5].setData("Binod", 80.0);
        students[6] = new Student();
        students[6].setData("Janak", 95.0);
        students[7] = new Student();
        students[7].setData("Gita", 89.0);
        students[8] = new Student();
        students[8].setData("Parbati", 91.0);
        students[9] = new Student();
        students[9].setData("Janaki", 87.0);
        System.out.println("Student Records:");
        for (Student s : students) {
            s.display();
        }
        Student topStudent = students[0];
        for (int i = 1; i < students.length; i++) {
            if (students[i].getPercentage() > topStudent.getPercentage()) {
                topStudent = students[i];
            }
        }
```

```
        System.out.println("\nStudent with Highest Average Marks:");
        System.out.println("Name: " + topStudent.getName() + ", Percentage: " + topStudent.getPercentage());
    }
}
```

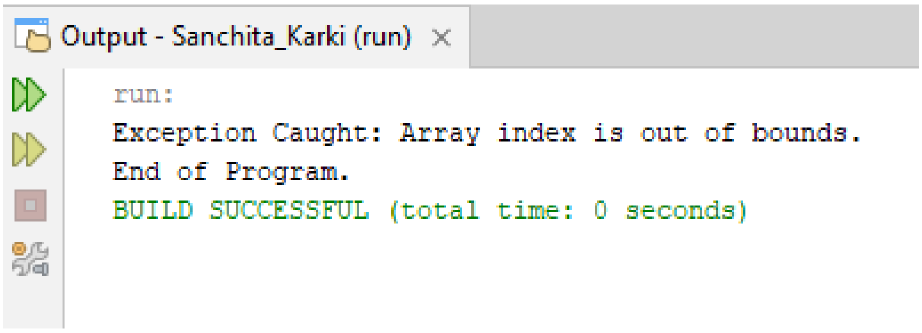


Output - Sanchita_Karki (run)

```
run:
Student Records:
Name: Sanchita, Percentage: 85.5
Name: Binu, Percentage: 90.0
Name: Ram, Percentage: 78.0
Name: Sita, Percentage: 88.5
Name: Shyam, Percentage: 92.0
Name: Binod, Percentage: 80.0
Name: Janak, Percentage: 95.0
Name: Gita, Percentage: 89.0
Name: Parbati, Percentage: 91.0
Name: Janaki, Percentage: 87.0

Student with Highest Average Marks:
Name: Janak, Percentage: 95.0
BUILD SUCCESSFUL (total time: 1 second)
```

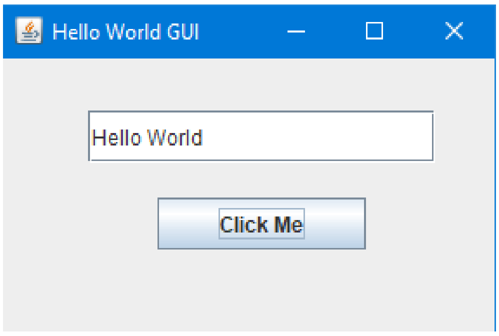
43. Write a program to illustrate the concept of ArrayIndexOutOfBoundsException

```
public class ArrayExceptionExample {
    public static void main(String[] args) {
        try {
            int[] arr = {10, 20, 30, 40, 50};
            System.out.println("Element: " + arr[10]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Exception Caught: Array index is out of bounds.");
        }
        System.out.println("End of Program.");
    }
}
```



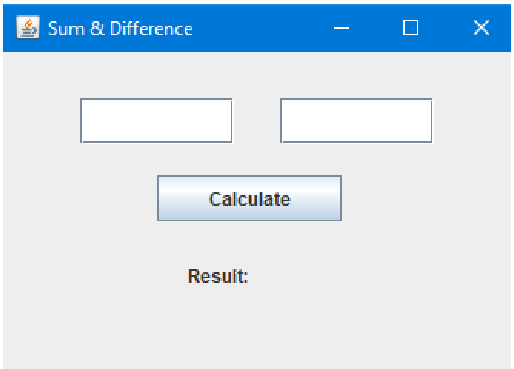
44. Write a Simple GUI program that displays “hello World” in a text field. The program should display if user clicks a button.

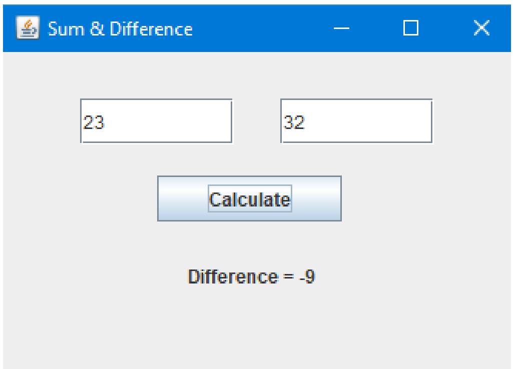
```
import javax.swing.*;
import java.awt.event.*;
public class HelloWorldGUI {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Hello World GUI");
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);
        JTextField textField = new JTextField();
        textField.setBounds(50, 30, 200, 30);
        frame.add(textField);
        JButton button = new JButton("Click Me");
        button.setBounds(90, 80, 120, 30);
        frame.add(button);
        button.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                textField.setText("Hello World");
            }
        });
        frame.setVisible(true);
    }
}
```

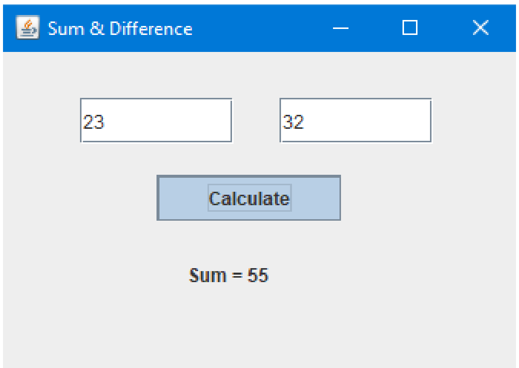


45. Write GUI program using Swing components to find sum and difference of two numbers. Use two text fields for giving input and a label for output. The program should display sum if user presses mouse and difference if user release mouse.

```
import javax.swing.*;
import java.awt.event.*;
public class SumDifferenceGUI {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Sum & Difference");
        frame.setSize(350, 250);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);
        JTextField t1 = new JTextField();
        t1.setBounds(50, 30, 100, 30);
        frame.add(t1);
        JTextField t2 = new JTextField();
        t2.setBounds(180, 30, 100, 30);
        frame.add(t2);
        JButton button = new JButton("Calculate");
        button.setBounds(100, 80, 120, 30);
        frame.add(button);
        JLabel resultLabel = new JLabel("Result:");
        resultLabel.setBounds(120, 130, 150, 30);
        frame.add(resultLabel);
        button.addMouseListener(new MouseAdapter() {
            public void mousePressed(MouseEvent e) {
                try {
                    int a = Integer.parseInt(t1.getText());
                    int b = Integer.parseInt(t2.getText());
                    int sum = a + b;
                    resultLabel.setText("Sum = " + sum);
                } catch (Exception ex) {
                    resultLabel.setText("Invalid Input");
                }
            }
            public void mouseReleased(MouseEvent e) {
                try {
                    int a = Integer.parseInt(t1.getText());
                    int b = Integer.parseInt(t2.getText());
                    int diff = a - b;
                    resultLabel.setText("Difference = " + diff);
                } catch (Exception ex) {
                    resultLabel.setText("Invalid Input");
                }
            }
        });

        // Show frame
        frame.setVisible(true);
    }
}
```


Sum & Difference
Calculate
Result:


Sum & Difference
23
32
Calculate
Difference = -9


Sum & Difference
23
32
Calculate
Sum = 55

46. You are hired by a reputed software company which is going to design an application for “Movie Rental System”. Your responsibility is to design a schema named MRS and create a table named Movie(id, Title, Genre, Language, Length). Write a program to design a GUI to take input for this table and insert the data into table after clicking OK button

Step 1: SQL Setup

```
CREATE DATABASE MRS;
USE MRS;
CREATE TABLE Movie (
   id INT PRIMARY KEY,
   Title VARCHAR(100),
   Genre VARCHAR(50),
   Language VARCHAR(50),
   Length INT
);
```

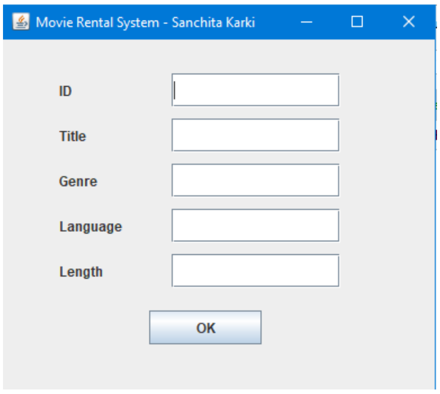
Step 2: Java Program

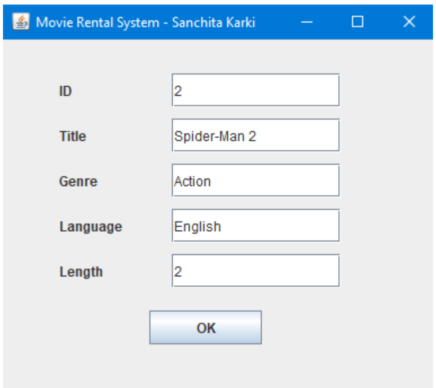
```
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;
public class MovieGUI {
   static final String URL = "jdbc:mysql://localhost:3306/MRS";
   static final String USER = "root";
   static final String PASS = "";
   public static void main(String[] args) {
      JFrame frame = new JFrame("Movie Rental System - Sanchita Karki");
      frame.setSize(400, 350);
      frame.setLayout(null);
      frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      JLabel l1 = new JLabel("ID");
      l1.setBounds(50, 30, 100, 30);
      frame.add(l1);
      JLabel l2 = new JLabel("Title");
      l2.setBounds(50, 70, 100, 30);
      frame.add(l2);
      JLabel l3 = new JLabel("Genre");
      l3.setBounds(50, 110, 100, 30);
      frame.add(l3);
      JLabel l4 = new JLabel("Language");
      l4.setBounds(50, 150, 100, 30);
      frame.add(l4);
      JLabel l5 = new JLabel("Length");
      l5.setBounds(50, 190, 100, 30);
      frame.add(l5);
      JTextField t1 = new JTextField();
      t1.setBounds(150, 30, 150, 30);
      frame.add(t1);
      JTextField t2 = new JTextField();
      t2.setBounds(150, 70, 150, 30);
      frame.add(t2);
      JTextField t3 = new JTextField();
      t3.setBounds(150, 110, 150, 30);
      frame.add(t3);
      JTextField t4 = new JTextField();
```

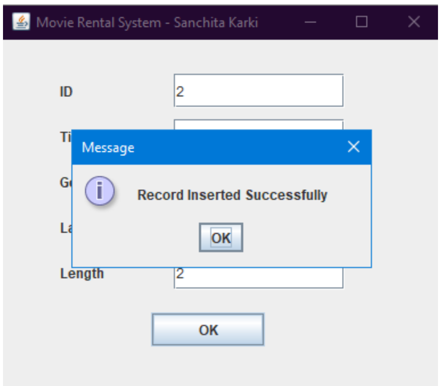
```
        t4.setBounds(150, 150, 150, 30);
        frame.add(t4);
        JTextField t5 = new JTextField();
        t5.setBounds(150, 190, 150, 30);
        frame.add(t5);
        JButton btn = new JButton("OK");
        btn.setBounds(130, 240, 100, 30);
        frame.add(btn);
        btn.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                try {
                    int id = Integer.parseInt(t1.getText());
                    String title = t2.getText();
                    String genre = t3.getText();
                    String language = t4.getText();
                    int length = Integer.parseInt(t5.getText());
                    Connection con = DriverManager.getConnection(URL, USER, PASS);
                    String query = "INSERT INTO Movie(id, Title, Genre, Language, Length)
VALUES(?,?,?,?,?)";
                    PreparedStatement pst = con.prepareStatement(query);
                    pst.setInt(1, id);
                    pst.setString(2, title);
                    pst.setString(3, genre);
                    pst.setString(4, language);
                    pst.setInt(5, length);
                    pst.executeUpdate();
                    JOptionPane.showMessageDialog(frame,
                            "Record Inserted Successfully");
                    con.close();
                } catch (Exception ex) {
                    JOptionPane.showMessageDialog(frame,
                            "Error : " + ex.getMessage());
                }
            }
        });
        frame.setVisible(true);
    }
}
```

Run SQL query/queries on server “127.0.0.1”:

```sql
CREATE DATABASE MRS;

USE MRS;

CREATE TABLE Movie(
    id INT PRIMARY KEY,
    Title VARCHAR(100),
    Genre VARCHAR(50),
    Language VARCHAR(50),
    Length INT
);
```







←T→	id	Title	Genre	Language	Length
Edit Copy Delete	1	Spider-Man	Action	English	2
Edit Copy Delete	2	Spider-Man 2	Action	English	2

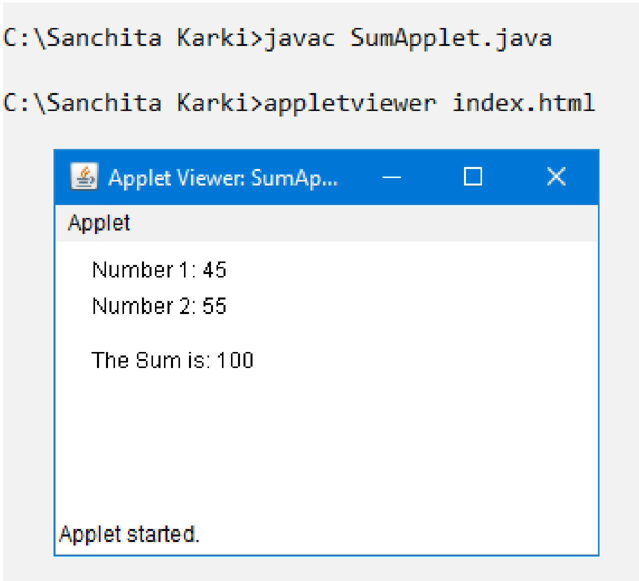
47. Write an applet that reads two numbers from HTML file as parameter and displays the sum of these two numbers.

Step 1: Java Program (SumApplet.java)

```
import java.applet.Applet;
import java.awt.Graphics;
public class SumApplet extends Applet {
   int sum;
   public void init() {
      String val1 = getParameter("num1");
      String val2 = getParameter("num2");
      int n1 = Integer.parseInt(val1);
      int n2 = Integer.parseInt(val2);
      sum = n1 + n2;
   }
   public void paint(Graphics g) {
      g.drawString("Number 1: " + getParameter("num1"), 20, 20);
      g.drawString("Number 2: " + getParameter("num2"), 20, 40);
      g.drawString("The Sum is: " + sum, 20, 70);
   }
}
```

Step 2: HTML File (index.html)

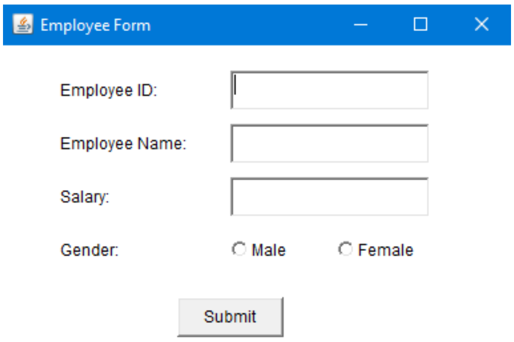
```
<html>
   <body>
      <applet code="SumApplet.class" width="300" height="150">
         <!-- Parameters passed to the applet -->
         <param name="num1" value="45">
         <param name="num2" value="55">
      </applet>
   </body>
</html>
```

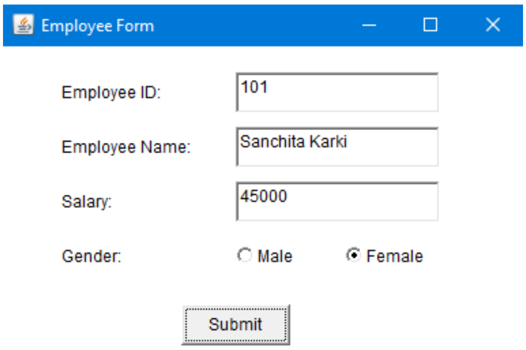


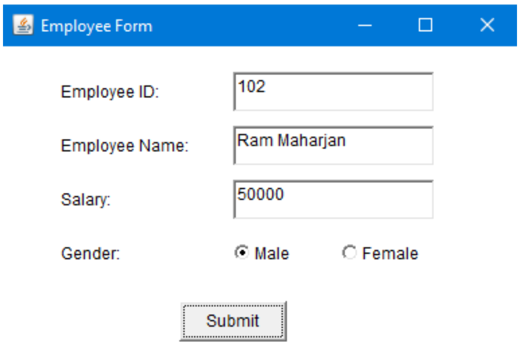
48. Write a Java program in awt to create form to enter employee information (eid, ename, salary, gender)

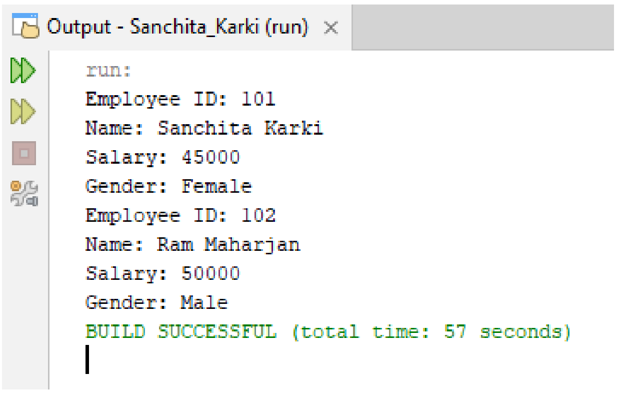
```
import java.awt.*;
import java.awt.event.*;
public class EmployeeForm extends Frame {
    Label l1, l2, l3, l4;
    TextField t1, t2, t3;
    Checkbox male, female;
    CheckboxGroup genderGroup;
    Button submit;
    public EmployeeForm() {
        setLayout(null);
        l1 = new Label("Employee ID:");
        l1.setBounds(50, 50, 100, 30);
        add(l1);
        l2 = new Label("Employee Name:");
        l2.setBounds(50, 90, 120, 30);
        add(l2);
        l3 = new Label("Salary:");
        l3.setBounds(50, 130, 100, 30);
        add(l3);
        l4 = new Label("Gender:");
        l4.setBounds(50, 170, 100, 30);
        add(l4);
        t1 = new TextField();
        t1.setBounds(180, 50, 150, 30);
        add(t1);
        t2 = new TextField();
        t2.setBounds(180, 90, 150, 30);
        add(t2);
        t3 = new TextField();
        t3.setBounds(180, 130, 150, 30);
        add(t3);
        genderGroup = new CheckboxGroup();
        male = new Checkbox("Male", genderGroup, false);
        male.setBounds(180, 170, 70, 30);
        add(male);
        female = new Checkbox("Female", genderGroup, false);
        female.setBounds(260, 170, 80, 30);
        add(female);
        submit = new Button("Submit");
        submit.setBounds(140, 220, 80, 30);
        add(submit);
        setTitle("Employee Form");
        setSize(400, 300);
        setVisible(true);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                dispose();
            }
        });
        submit.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                String eid = t1.getText();
                String ename = t2.getText();
```

```
            String salary = t3.getText();
            String gender = genderGroup.getSelectedCheckbox() != null
                ? genderGroup.getSelectedCheckbox().getLabel()
                : "Not selected";
            System.out.println("Employee ID: " + eid);
            System.out.println("Name: " + ename);
            System.out.println("Salary: " + salary);
            System.out.println("Gender: " + gender);
          }
        });
    }
    public static void main(String[] args) {
        new EmployeeForm();
    }
}
```



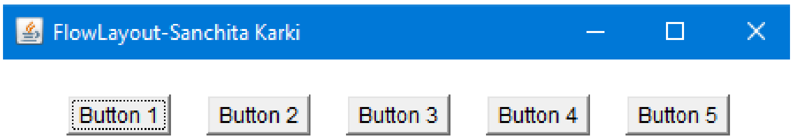


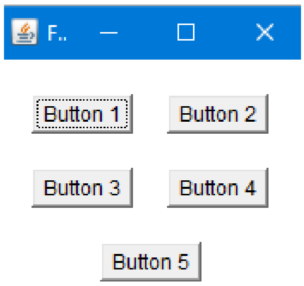


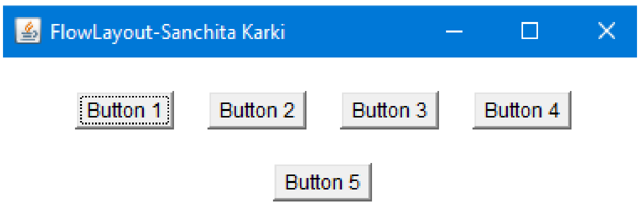


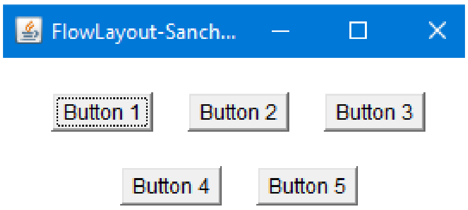
49. Write a Java program to demonstrate FlowLayout

```
import java.awt.*;
import java.awt.event.*;
public class FlowLayoutDemo extends Frame {
    public FlowLayoutDemo() {
        setLayout(new FlowLayout(FlowLayout.CENTER, 20, 20));
        add(new Button("Button 1"));
        add(new Button("Button 2"));
        add(new Button("Button 3"));
        add(new Button("Button 4"));
        add(new Button("Button 5"));
        setTitle("FlowLayout-Sanchita Karki");
        setSize(400, 200);
        setVisible(true);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                dispose();
            }
        });
    }
    public static void main(String[] args) {
        new FlowLayoutDemo();
    }
}
```



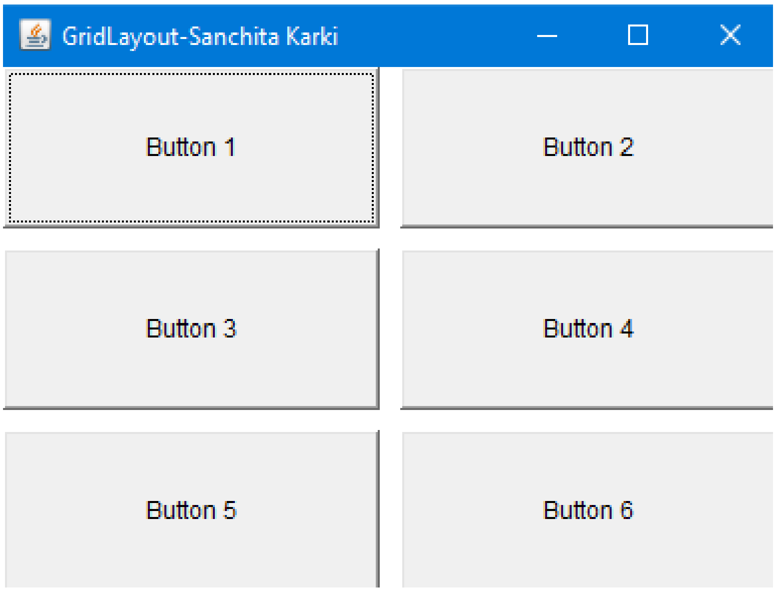






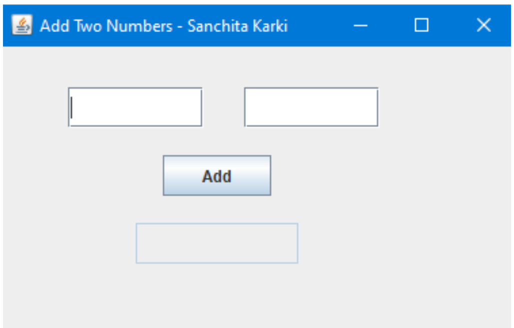
50. Write a Java Program to demonstrate GridLayout

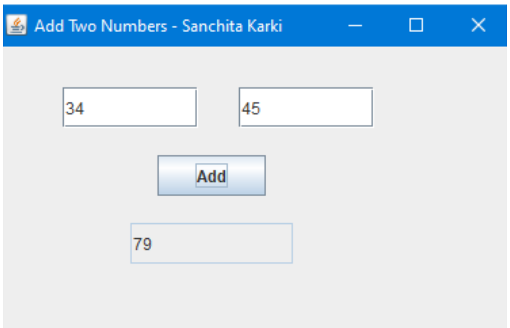
```
import java.awt.*;
import java.awt.event.*;
public class GridLayoutDemo extends Frame {
    public GridLayoutDemo() {
        setLayout(new GridLayout(3, 2, 10, 10));
        add(new Button("Button 1"));
        add(new Button("Button 2"));
        add(new Button("Button 3"));
        add(new Button("Button 4"));
        add(new Button("Button 5"));
        add(new Button("Button 6"));
        setTitle("GridLayout-Sanchita Karki");
        setSize(400, 300);
        setVisible(true);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                dispose();
            }
        });
    }
    public static void main(String[] args) {
        new GridLayoutDemo();
    }
}
```



51. Write a program using swing components to add two numbers. Use text fields for inputs and output. Your program should display the result when the user presses a button.

```
import javax.swing.*;
import java.awt.event.*;
public class AddTwoNumbers {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Add Two Numbers - Sanchita Karki");
        frame.setSize(350, 250);
        frame.setLayout(null);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JTextField t1 = new JTextField();
        t1.setBounds(50, 30, 100, 30);
        frame.add(t1);
        JTextField t2 = new JTextField();
        t2.setBounds(180, 30, 100, 30);
        frame.add(t2);
        JButton button = new JButton("Add");
        button.setBounds(120, 80, 80, 30);
        frame.add(button);
        JTextField result = new JTextField();
        result.setBounds(100, 130, 120, 30);
        result.setEditable(false);
        frame.add(result);
        button.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                try {
                    int a = Integer.parseInt(t1.getText());
                    int b = Integer.parseInt(t2.getText());
                    int sum = a + b;
                    result.setText(String.valueOf(sum));
                } catch (Exception ex) {
                    result.setText("Invalid Input");
                }
            }
        });
        frame.setVisible(true);
    }
}
```

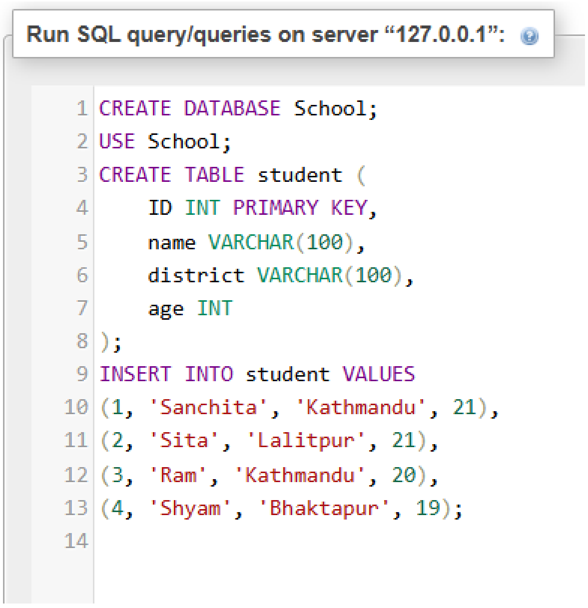




52. Write a Java program who live in Kathmandu district, assuming that the student table has four attributes (ID, name, district and age).

Step 1: SQL Setup

```
CREATE DATABASE School;
USE School;
CREATE TABLE student (
   ID INT PRIMARY KEY,
   name VARCHAR(100),
   district VARCHAR(100),
   age INT
);
INSERT INTO student VALUES
(1, 'Sanchita', 'Kathmandu', 21),
(2, 'Sita', 'Lalitpur', 21),
(3, 'Ram', 'Kathmandu', 20),
(4, 'Shyam', 'Bhaktapur', 19);
```


Run SQL query/queries on server “127.0.0.1”:

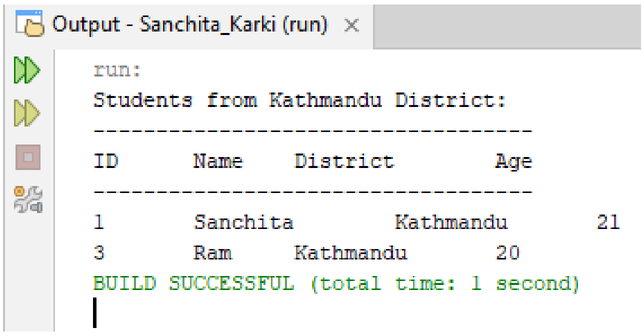
```
CREATE DATABASE School;
USE School;
CREATE TABLE student (
    ID INT PRIMARY KEY,
    name VARCHAR(100),
    district VARCHAR(100),
    age INT
);
INSERT INTO student VALUES
(1, 'Sanchita', 'Kathmandu', 21),
(2, 'Sita', 'Lalitpur', 21),
(3, 'Ram', 'Kathmandu', 20),
(4, 'Shyam', 'Bhaktapur', 19);
```

ID	name	district	age
1	Sanchita	Kathmandu	21
2	Sita	Lalitpur	21
3	Ram	Kathmandu	20
4	Shyam	Bhaktapur	19

Step 2: Java Program

```
import java.sql.*;
public class KathmanduStudents {
   public static void main(String[] args) {
      String url = "jdbc:mysql://localhost:3306/School";
      String user = "root";
      String password = "";
      try {
         Class.forName("com.mysql.cj.jdbc.Driver");
         Connection con = DriverManager.getConnection(url, user, password);
```

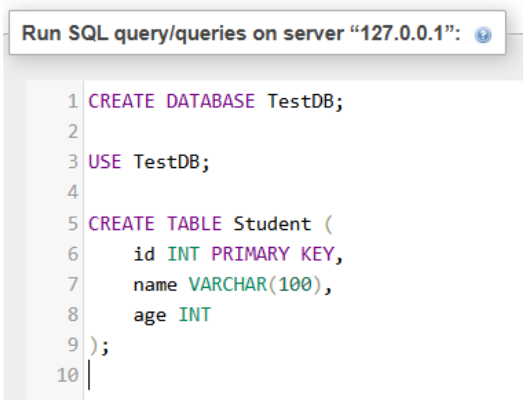
```
            String query = "SELECT * FROM student WHERE district = 'Kathmandu'";
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(query);
            System.out.println("Students from Kathmandu District:");
            System.out.println("-----------------------------------");
            System.out.println("ID\tName\tDistrict\tAge");
            System.out.println("-----------------------------------");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("ID") + "\t" +
                    rs.getString("name") + "\t" +
                    rs.getString("district") + "\t" +
                    rs.getInt("age")
                );
            }
            con.close();

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```



53. Write a Java program to insert one record to database. Assume your own database and table

Step 1: SQL Setup

```
CREATE DATABASE TestDB;
USE TestDB;
CREATE TABLE Student (
   id INT PRIMARY KEY,
   name VARCHAR(100),
   age INT
);
```

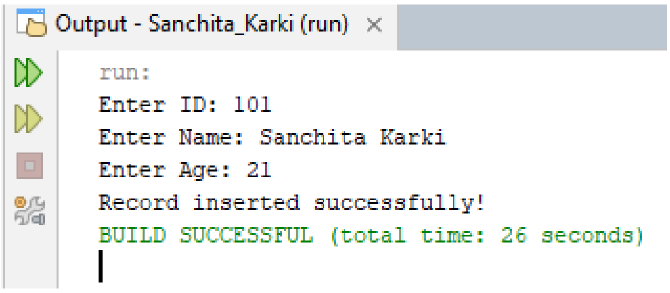



```
CREATE DATABASE TestDB;

USE TestDB;

CREATE TABLE Student (
    id INT PRIMARY KEY,
    name VARCHAR(100),
    age INT
);
```

Step 2: Java Program

```
import java.sql.*;
import java.util.Scanner;
public class InsertRecord {
   public static void main(String[] args) {
      String url = "jdbc:mysql://localhost:3306/TestDB";
      String user = "root";
      String password = "";
      Scanner sc = new Scanner(System.in);
      try {
         System.out.print("Enter ID: ");
         int id = sc.nextInt();
         sc.nextLine();
         System.out.print("Enter Name: ");
         String name = sc.nextLine();
         System.out.print("Enter Age: ");
         int age = sc.nextInt();
         Connection con = DriverManager.getConnection(url, user, password);
         String query = "INSERT INTO Student (id, name, age) VALUES (?, ?, ?)";
         PreparedStatement pst = con.prepareStatement(query);
         pst.setInt(1, id);
         pst.setString(2, name);
         pst.setInt(3, age);
         int result = pst.executeUpdate();
         if (result > 0) {
            System.out.println("Record inserted successfully!");
         } else {
            System.out.println("Insert failed!");
         }
         con.close();
```

```
            sc.close();
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```



id	name	age
101	Sanchita Karki	21

54. Write a Java Program to delete a record from database. Assume your own database and table

Step 1: SQL Setup
Use the database created for previous question.

ID	name	district	age
1	Sanchita	Kathmandu	21
2	Sita	Lalitpur	21
3	Ram	Kathmandu	20
4	Shyam	Bhaktapur	19

Step 2: Java Program

```
import java.sql.*;
import java.util.Scanner;
public class DeleteRecord {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/school";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            System.out.print("Enter ID to delete: ");
            int id = sc.nextInt();
            Connection con = DriverManager.getConnection(url, user, password);
            String query = "DELETE FROM Student WHERE id = ?";
            PreparedStatement pst = con.prepareStatement(query);
            pst.setInt(1, id);
            int result = pst.executeUpdate();
            if (result > 0) {
                System.out.println("Record deleted successfully!");
            } else {
                System.out.println("Record not found!");
            }
            con.close();
            sc.close();
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

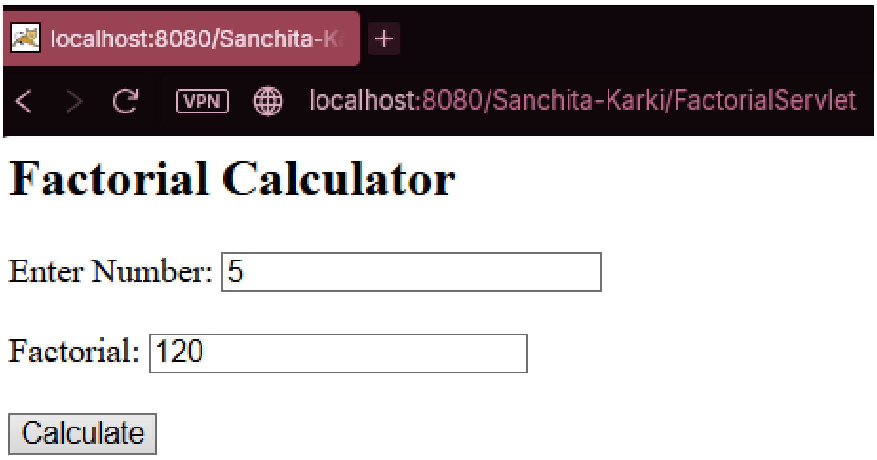
Output - Sanchita_Karki (run)

```
run:
Enter ID to delete: 3
Record deleted successfully!
BUILD SUCCESSFUL (total time: 5 seconds)
```

ID	name	district	age
1	Sanchita	Kathmandu	21
2	Sita	Lalitpur	21
4	Shyam	Bhaktapur	19

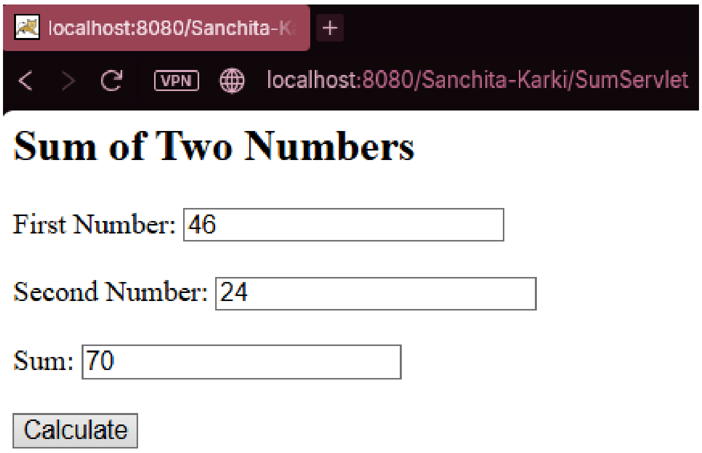
55. Create a servlet that displays two text boxes in web browser, reads number entered in first text box, calculates factorial and displays it in second textfield.

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;
@WebServlet("/FactorialServlet")
public class FactorialServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String numStr = request.getParameter("num");
      String result = "";
      if (numStr != null && !numStr.equals("")) {
         int n = Integer.parseInt(numStr);
         int fact = 1;
         for (int i = 1; i <= n; i++) {
            fact = fact * i;
         }
         result = String.valueOf(fact);
      }
      out.println("<html><body>");
      out.println("<h2>Factorial Calculator</h2>");
      out.println("<form action='FactorialServlet' method='GET'>");
      out.println("Enter Number: ");
      out.println("<input type='text' name='num' value='" +
            (numStr != null ? numStr : "") + "'/>");
      out.println("<br><br>");
      out.println("Factorial: ");
      out.println("<input type='text' name='res' value='" + result + "' readonly/>");
      out.println("<br><br>");
      out.println("<input type='submit' value='Calculate'/>");
      out.println("</form>");
      out.println("</body></html>");
   }
}
```



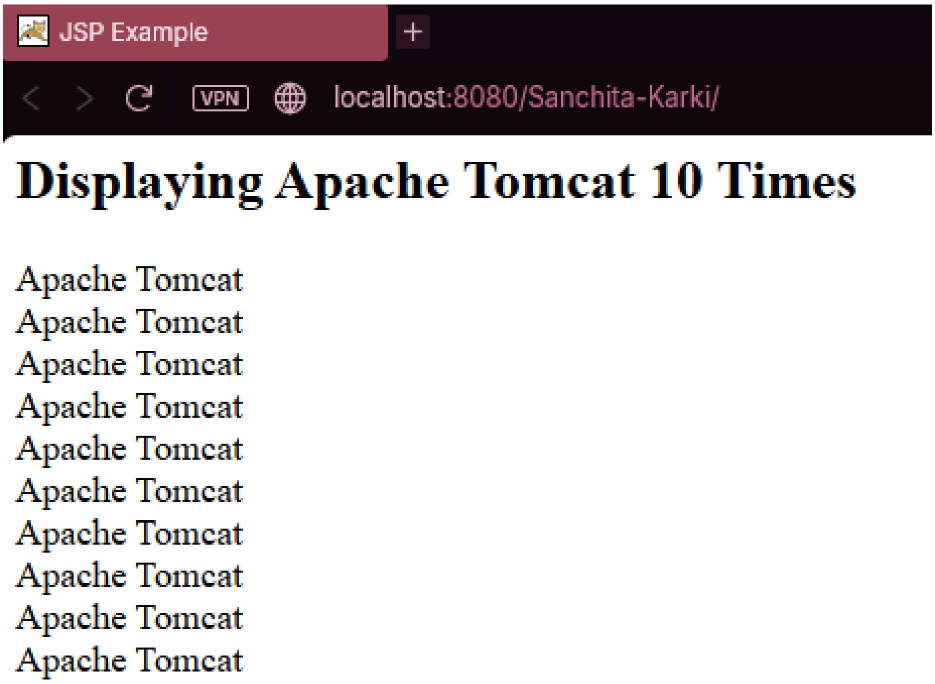
56. Write a servlet program that reads two numbers from web browser and finds sum of these two numbers.

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;
@WebServlet("/SumServlet")
public class SumServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String n1 = request.getParameter("num1");
        String n2 = request.getParameter("num2");
        int sum = 0;
        if (n1 != null && n2 != null &&
                !n1.equals("") && !n2.equals("")) {
            int a = Integer.parseInt(n1);
            int b = Integer.parseInt(n2);
            sum = a + b;
        }
        out.println("<html><body>");
        out.println("<h2>Sum of Two Numbers</h2>");
        out.println("<form action='SumServlet' method='GET'>");
        out.println("First Number: ");
        out.println("<input type='text' name='num1' value='" +
                (n1 != null ? n1 : "") + "'/>");
        out.println("<br><br>");
        out.println("Second Number: ");
        out.println("<input type='text' name='num2' value='" +
                (n2 != null ? n2 : "") + "'/>");
        out.println("<br><br>");
        out.println("Sum: ");
        out.println("<input type='text' value='" + sum + "' readonly/>");
        out.println("<br><br>");
        out.println("<input type='submit' value='Calculate'/>");
        out.println("</form>");
        out.println("</body></html>");
    }
}
```



57. Write a JSP program display text “Apache Tomcat” 10 times.

```
<%@ page contentType="text/html" pageEncoding="UTF-8" %>
<html>
<head>
   <title>JSP Example</title>
</head>
<body>
<h2>Displaying Apache Tomcat 10 Times</h2>
<%
   for(int i = 1; i <= 10; i++)
   {
%>
Apache Tomcat <br>
<%
   }
%>
</body>
</html>
```



58. How exceptions can be handled in JSP scripts? Explain with suitable JSP script

Exceptions in JSP can be handled using:

1. try-catch block
2. JSP errorPage directive
3. JSP isErrorPage directive

Exception handling prevents abnormal termination of JSP pages and displays user-friendly error messages.

Using try-catch:

```
<%@ page contentType="text/html" pageEncoding="UTF-8" %>
<html>
<head>
   <title>Exception Handling</title>
</head>
<body>
<h3>Exception Handling using try-catch</h3>
<%
   try
   {
      int a = 10;
      int b = 0;
      int c = a / b;
      out.println("Result = " + c);
   }
   catch(Exception e)
   {
      out.println("Exception Occurred : " + e.getMessage());
   }
%>
</body>
</html>
```